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Mapping the Cellular Biogeography of Human Bone Marrow Niches Using Single-Cell Transcriptomics and Proteomic Imaging

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KT, LQ, and ShB conceptualized and designed the project. ShB, KJA, MD, AT, ChiC, and YL performed the data generation and experiments. ShB, KJA, MD, JS, MP, DS, GD, JH, IZ, YL, BX, and TI performed analysis and wrote software and made figures. ShB, KJA, MD, JS, ChaC, JX, MR, ZM, CD, VP, OS, DO, IM, MC, CN, LQ, and KT provided critical conceptual contributions and technical expertise. CN, SiB, OS, and CC assisted with reagent validation and provided clinical samples. ShB, KJA, MD, LQ, and KT wrote the manuscript.

Declaration of Interests

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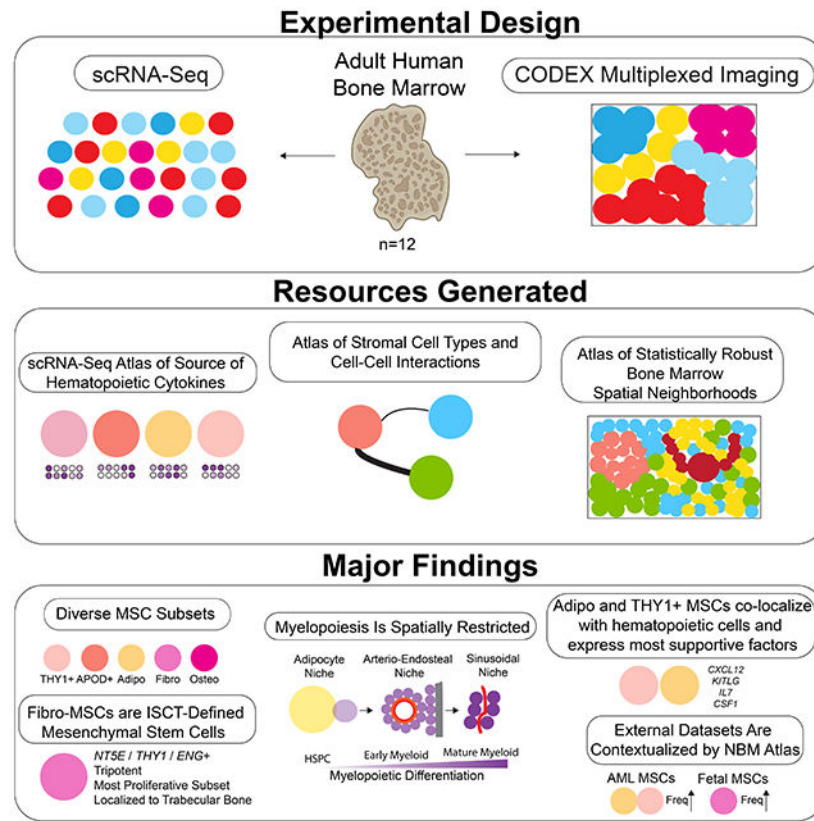
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Summary

Non-hematopoietic cells are essential contributors to hematopoiesis. However, heterogeneity and spatial organization of these cells in human bone marrow remain largely uncharacterized. We used single-cell RNA sequencing to profile 29,325 non-hematopoietic cells and discovered nine transcriptionally distinct subtypes. We simultaneously profiled 53,417 hematopoietic cells and predicted their interactions with non-hematopoietic subsets. We employed Co-Detection by Indexing (CODEX) to spatially profile over 1.2 million cells. We integrated scRNA-Seq and CODEX data to link predicted cellular signaling with spatial proximity. Our analysis revealed a hyperoxygenated arterio-endosteal neighborhood for early myelopoiesis, and an adipocytic localization for early hematopoietic stem and progenitor cells. We used our CODEX atlas to annotate new images and uncovered mesenchymal stromal cell (MSC) expansion and spatial neighborhoods co-enriched for leukemic blasts and MSCs in AML patient samples. This spatially-resolved, multiomic atlas of human bone marrow provides a reference for investigation of cellular interactions that drive hematopoiesis.

Graphical Abstract



Introduction

The bone marrow is a complex organ which houses diverse cells from the hematopoietic, mesenchymal, endothelial, vascular smooth muscle, and neural lineages¹. Relatively rare non-hematopoietic cells are known to make critical contributions to hematopoiesis². Many non-hematopoietic cell types, including endothelial cells, mesenchymal stromal cells (MSCs), and osteoblasts have been proposed to be critical constituents of the bone marrow niche³⁻⁶. Recent technological advances employing single-cell RNA sequencing (scRNA-Seq) revealed the existence of multiple subpopulations of these non-hematopoietic marrow elements in mice⁷⁻¹⁰.

Despite extensive work studying human hematopoietic cells, there remains a relative paucity of analogous studies defining the non-hematopoietic cells that make up the human bone marrow microenvironment¹¹⁻¹³. Defining the single-cell composition of bone marrow niches has been limited by challenges in isolating sufficient viable non-hematopoietic cells, which make up <0.5% of the marrow cellularity^{14,15}. Moreover, specific cell isolation methods may greatly influence the diversity of cell types captured, as shown in mice⁹. Bone marrow aspiration, analogous to flushing of mouse bones, fails to capture cells which are tightly adhered to the bone surface and bias the bone marrow MSC composition towards adipocytic, rather than osteogenic cells. Much of human MSC profiling has been performed on such aspirate samples^{13,16-18}, which have indeed yielded mostly adipogenic MSCs. Other human MSC profiling has been performed following a period of *ex vivo* expansion^{19,20}, which may

alter the transcriptional profile of MSCs from their *in vivo* homeostatic state. Furthermore, most aforementioned analyses lack spatial information of these niche cells, and it remains unknown whether there are human MSC subtypes with distinct spatial organization.

Here, we overcame these limitations by performing transcriptomic profiling of single cells from enzymatically digested fresh human bone marrow tissue without the need for expansion. We employed scRNA-Seq to comprehensively characterize the transcriptomic profiles of both hematopoietic and non-hematopoietic cell types, predict patterns of intercellular communication, and identify the mediating signaling molecules. In parallel, we used Co-Detection by Indexing (CODEX)²¹ with a 53-antibody panel to systematically and simultaneously characterize all major bone marrow cell types and microenvironmental structures. We uncovered a significant non-hematopoietic diversity in the human bone marrow with discrete spatial phenotypes and quantitatively defined how niche elements are organized. Furthermore, we demonstrated the utility of our CODEX atlas in acute myeloid leukemia (AML) patients, where we uncovered enriched MSC-leukemic blast spatial interactions. Collectively, this work represents the most comprehensive, spatially-resolved, multiomic human bone marrow atlas which will serve as an important reference for future studies of the human bone marrow niche.

Results

A Comprehensive scRNA-Seq Atlas of the Human Bone Marrow

To uncover the cellular composition of the human bone marrow, we developed an experimental pipeline for enzymatic release of cells from femoral head tissue (Figures S1A-B). We confirmed that areas of interest had normal trabecular structure by performing micro-computed tomography analysis (trabecular bone volume fraction: 0.261 $\pm$ 0.098 (mean $\pm$ SD), n=6, Figure S1A). Next, we devised an enrichment strategy that would capture hematopoietic cells, rare hematopoietic stem and progenitor cells (HSPCs), and non-hematopoietic microenvironmental cells (Figure 1A, STAR Methods). The three populations were pooled in a ratio favoring representation of rare HSPCs and non-hematopoietic cells and then subjected to scRNA-Seq (Figure 1A).

We collected bone marrow from fresh femoral heads from 16 individuals between 52-80 years of age who underwent hip replacement surgery, performing scRNA-Seq on 12 of them and CODEX on another 12, with 8 overlapping samples (Table S1). In total, we sequenced 53,417 hematopoietic cells and 29,325 non-hematopoietic cells (Figures 1B-C). We captured rich, high-quality transcriptional information in each cell lineage, with per-cell medians of 16,903 UMIs, 3,117 unique genes, and ~3% mitochondrial reads detected after quality control filtering (Figure S1C).

We first analyzed the hematopoietic populations. As expected, we identified tri-lineage hematopoiesis and HSPC subsets (Figure 1D, Table S2). Additionally, we improved upon other hematopoiesis-focused bone marrow scRNA-Seq references^{11,12} by capturing the full granulocyte differentiation trajectory by virtue of our sample processing not including density-based isolation of mononuclear cells – a standard clinical tissue banking process which results in granulocyte loss due to their multi-lobated nuclei (*CSF3R*, Figures 1B,

S1D). Numerous plasma cells were also captured as a byproduct of the CD45 depletion procedure (*MZB1*, Figures 1B, S1D). Age did not skew the frequencies of captured cell types (Figure S1E).

Next, we analyzed the non-hematopoietic cell compartment. We identified three major cell lineages – vascular smooth muscle cells (*ACTA2*, *RGS5*, *TAGLN*), endothelial cells (*CDH5*, *PECAM1*, *VWF*), and mesenchymal cells (*CXCL12*, *PDGFRA*) including mesenchymal stromal cells (MSCs) and osteoblasts (Figures 1B-D, S1D, Table S2). Our analysis revealed far greater cellular diversity within these cells in bone marrow than existing scRNA-Seq references (Figure S1F). Notably, contrary to earlier findings that characterize mesenchymal stromal cells as relatively homogeneous with an adipocytic transcriptional profile^{13,16-18}, we found these enzymatically released cells to be highly heterogeneous with numerous clusters identified (Figures 1B, S1F).

Importantly, we did not observe an inflammatory response signature, except in cell types which express inflammatory cytokines as part of their identity (mature neutrophils and monocytes, e.g., *CXCL8* and *IL1B*), supporting the notion that these samples represent healthy aged bone marrow (Figures S2A-B). Moreover, compared to existing healthy donor aspirate-derived bone marrows, our samples did not express more inflammatory cytokines (Figure S2C). To further ensure our samples were not inflamed, we compared our data to published scRNA-Seq data from cultured human MSCs treated with TNF α and IFN γ , and found our mesenchymal cells had significantly less of an inflamed signature than known inflamed MSCs and were comparable to healthy aspirate MSCs from *Li et al* (Figure S2D).

Non-Hematopoietic Cell Subset Analysis Demonstrates MSC and Endothelial Cell Transcriptional Diversity

We discerned significant heterogeneity among mesenchymal cells, identifying osteo-lineage (*NCAM1*, *SPP1*, *BGLAP*), adipo-lineage (*APOE*, *LPL*, *PPARG*, *CEBPA*), and fibroblastic cells (*PDPN*, *CSPG4*, *DCN*, *DPT*) (Figures 2A-B, S3A-B). Osteolineage cells were divided into *IBSP/BGLAP*-high osteoblasts and *IBSP/BGLAP*-low Osteo-MSCs (Figure 2B). Despite not capturing adipocytes likely due to sample processing loss, we found a cell cluster with the highest *CXCL12* level and high expression of adipose lineage genes such as *CEBPA*, *PPARG*, *APOE*, and *LPL* which we labeled as Adipo-MSCs (Figures 2B, S3B). These cells resemble reported adipogenic precursors (MALPs)⁸ and Adipo-Cxcl12-abundant-reticular (Adipo-CAR) cells⁹ in mice. We also found a population of *THY1*+ MSCs which expressed adipocytic genes strongly, but also a unique expression profile including *THY1* and *LBP*, but lower *LPL* expression compared to Adipo-MSCs, which we termed *THY1*+ MSCs (Figures 2B, S3B). *THY1*+ MSCs did not have a clear murine counterpart²². Fibroblast-like cells that expressed *CXCL12*, highly expressed *DPT* and *NT5E*, and very specifically expressed *PDPN* and *HAS1* were also observed, which we termed Fibro-MSCs (Figure 2B). We also identified an *APOD*⁺ *GSM*^{hi} population of cells which we were unable to correlate to previously known cell types in humans or mice, and so we named them *APOD*+ MSCs as this gene was highly and specifically expressed (Figures 2A-B).

The Fibro-MSC population aligns closely with the previously described Early Mesenchymal Progenitor population⁸ in mice defined by *CPSG4*, *CD34*, and *DPT* expression and the human skeletal stem cell population²³ defined by *PDPN*, *CD164*, and *NT5E* (CD73) expression and lack of lineage marker expression (Figures 2B, S3B). Fibro-MSCs also expressed mesenchymal stem cell markers defined by the International Society for Cellular Therapy²⁴ (ISCT), namely *NT5E* (CD73), *THY1* (CD90), and *ENG* (CD105) (Figure 2C). It is important to note that expression of these three genes was highly variable among MSC subsets, and this bias should be considered when using them to isolate MSCs. In particular, inclusion of CD90 (*THY1*) as a sorting marker may bias the isolated MSC population towards Fibro and THY1+ MSCs. *NGFR* (*CD271*) was more consistently expressed among subsets, providing rationale for its continued use²⁵ as an unbiased MSC marker (Figure 2C). *MCAM* (*CD146*), however, was not highly expressed. *NES*, a MSC marker in mice²⁶, was not detected in human MSCs but was expressed in sinusoidal endothelial cells (SECs), arterial endothelial cells (AECs), and vascular smooth muscle cells (VSMCs) (Figure S3B). It should be noted that there was considerable variability in mesenchymal cell frequency between samples, which could reflect dynamic cell states such as activation or technical factors including sampling differences (Figure S3C).

Using CytoTRACE pseudotime analysis²⁷, we determined that Fibro-MSCs were the most primitive among MSC subsets, consistent with their transcriptomic similarity to reported mesenchymal progenitors and the ISCT definition (Figures 2C-D). THY1+ MSCs also had a high CytoTRACE score, suggesting that they may also be more primitive. To validate that Fibro- and THY1+ MSCs were more stem/progenitor-like, we sorted four MSC subpopulations (Adipo-MSCs, THY1+ MSCs, Osteo-MSCs/Osteoblasts, and Fibro-MSCs) based on markers derived from our scRNA-Seq data (Figures S3D-E) and performed fibroblast colony forming unit (CFU-F) assays. Sorted Podoplanin+ Fibro-MSCs exhibited the highest CFU-F forming capacity, followed by THY1+ MSCs, with Osteo-MSCs and Adipo-MSCs showing the least (Figure 2E). Cultured Fibro- and THY1+ MSCs maintained proliferation at high passages while Adipo-MSCs and Osteo-MSCs slowed progressively (Figure 2F). Fibro-MSCs could differentiate into osteoblasts, adipocytes, and chondrocytes after being cultured in the appropriate differentiation medium (Figures S3F-G). Notably, these experiments using flow sorted cells also confirmed the protein-level divergence of these MSC subsets and identified Podoplanin as a robust marker for mesenchymal stem cells. Overall, our findings underscore Fibro-MSCs' alignment with the human mesenchymal stem cell criteria defined by the ISCT and resemblance to reported early mesenchymal progenitors in both human and mouse^{8,23}.

Our data also captured two major classes of bone marrow endothelial cells – AECs and SECs (Figure 2G). AECs showed higher expression of *CXCL12*, *ICAM2*, and *PODXL*, but lower levels of venous genes such as *EPHB4* and *NR2F2* than SECs, which expressed more genes associated with endothelial-hematopoietic cell interactions (e.g., *ACKR1*, *SELE*) (Figure 2H). We did not find cells associated with Type H vessels that connect arterioles to sinusoids, possibly due to their low frequency and specific association with the growth plate of young individuals²⁸. Additionally, we did not capture a cluster of *PROX1+ PDPN+ LYVE1+* lymphatic endothelial cells but did observe rare *LYVE1*-expressing cells (Figure

S4A), which could be related to very rare bone marrow lymphatic progenitors previously described^{29,30}.

To further emphasize the importance of capturing the heterogeneity of non-hematopoietic cells, we mapped multiple published scRNA-Seq datasets of healthy human MSCs^{13,16-18,31,32} to our atlas. We observed a dominance of Adipo-MSCs in aspirate-derived samples (Figure 2I). Our atlas captured 24.4-fold more non-Adipo-MSC subsets than all mapped adult datasets combined (Figure 2I). Fibro-MSCs were essentially not captured at all in other adult datasets. Only 76 THY1+ MSCs were captured across all other datasets combined, compared to 4,178 in our dataset (Figure 2I). Since Jardine *et al.* was the only other non-aspirate derived dataset (crushed fetal femur), we were interested to compare fetal cell states to adult cell states. Most fetal MSC populations corresponded well with adult MSC populations, with mapping of labels from Jardine *et al.* to corresponding labels in our atlas -- “fibroblasts” to Fibro-MSCs, “Adipo-CAR” to Adipo and THY1+ MSCs, and “osteogenic populations” to osteo-MSCs/osteoblasts/APOD+ MSCs (Figure S4B). We next examined the differences between fetal and adult Fibro-MSCs, finding that fetal Fibro-MSCs had a radial glial-like signature perhaps indicative of neural crest origin³³ and that adult Fibro-MSCs more closely resemble prototypical fibroblasts (Figure S4C). In sum, we found that there is previously unrepresented and functionally consequential diversity within human bone marrow MSCs that cannot be captured efficiently from aspirate samples and demonstrated the utility of our atlas as a healthy reference to characterize similarities and differences to MSCs in other scRNA-Seq datasets.

MSCs, Endothelial Cells, and Osteolineage Cells Cooperate to Produce Diverse Hematopoietic Support Factors

We next asked whether MSC heterogeneity translated to differential transcript levels of support factors for hematopoiesis. Starting with hematopoietic factors which have been widely studied in non-hematopoietic cell subsets^{2,34,35} (Figure 3A), we found different MSC types specialized in producing specific support factors; for example, *CXCL12* was highest in Adipo- and THY1+ MSCs, while *IGF1* was predominantly expressed by Fibro-MSCs.

We next employed CellChat³⁶ to systematically interrogate cellular communications in the bone marrow based on cognate ligand and receptor co-expression between two cell types, as demonstrated by *CXCL12* and *SELE* interactions of MSCs and SECs with different HSPC subsets respectively (Figure 3B). To validate the predicted interactions between HSPCs and MSC subsets, we performed co-culture of freshly sorted MSC subsets and freshly isolated CD34+ cells. We found that sorted Adipo-, Osteo-, and THY1+ MSCs significantly preserved primitive Lin- CD34+ CD38- cells, along with a strong trend from Fibro-MSCs, as opposed to monoculture where these cells were scantily preserved (Figures 3C, S4D). We also found that globally, mesenchymal lineage cells had the most outgoing signaling contributions (i.e., significant interactions in which they expressed the ligand) (Figure 3D, Table S3).

Next, we interrogated support factor expression and predicted communication of non-hematopoietic cells with hematopoietic cells focusing on HSPC maintenance, myelopoiesis, and lymphopoiesis. We grouped support factors based on their shared pathway annotation

described in the CellChat database³⁶ (e.g. ANGPT family contains multiple Angiopoietin genes). Several cell types have been reported to be important for HSPC maintenance, including osteoblasts^{5,37,38} and perivascular MSCs^{3,4}. Our data suggests that both osteoblasts and MSCs may support human HSPCs, but most of the canonical support factors are produced by MSC subsets, especially Adipo- and THY1+ MSCs. For instance, Adipo- and THY1+ MSCs produced highest levels of canonical HSPC niche factors such as *CXCL12*³⁹ (CXCL family) and *KITLG*⁴⁰ (KIT family) which were broadly received by hematopoietic cells without strong specificity for HSPCs (Figures 3A,3E-F). Osteo-lineage cells expressed *TGFB1* and *CDH2* more than Adipo- or THY1+ MSCs (Figure 3A), which have been shown to support HSC quiescence in mice^{5,41}. HSCs also had significant predicted incoming signaling from E-Selectin provided by SECs (Figures 3A, 3F).

We next focused on factors known to be important for myelopoiesis. *CSF1* was expressed by multiple MSC subsets, with highest expression in Adipo-MSCs (Figure 3A). *CSF2* and *CSF3*, however, were not detected at high levels, although a subset of SECs (5.8%) did have detectable expression of *CSF3* (n.s., Figure S4E). This result may reflect the inherent challenge of detecting transient cytokine gene expression⁴². While most hematopoietic support factors were provided by mesenchymal cells, we also noted that CD4+ T cells, which are not typically thought of as critical niche cells, were major producers of *FLT3LG* in the bone marrow (Figure 3E, Table S3). HSCs, GMPs, and common lymphoid progenitors (CLPs) were predicted to receive *FLT3LG*, which is known to be critical for both myelopoiesis⁴³ and lymphopoiesis⁴⁴ (Figure 3F).

We then studied niche contributions to lymphopoiesis by interrogating factors such as *IL7* and Notch ligands^{45,46}. *IL7* was primarily produced by Adipo- and THY1+ MSCs (Figure 3A). The concept that the vascular niche provides the majority of Notch signaling in the bone marrow has been suggested in mice⁷, while in humans CD146+ MSCs⁴⁷ have been proposed to be critical and the vascular contribution has not been clearly delineated. Collectively, endothelial cells and VSMCs are indeed the major contributors of *DLL1*, *DLL4*, *JAG1*, and *JAG2* in the human bone marrow, with arterioles (AECs and VSMCs) expressing the most *JAG1* (Figure 3A). We also noted that Notch signaling was predicted as outgoing from mesenchymal cells (Figure 3E) but was weaker than endothelial-driven Notch signaling and largely based on *JAG1* expression. Interestingly, VSMCs also received significant incoming Notch signaling through *NOTCH3*, possibly received from AECs (Figure 3F, Table S3). Notably, all of these non-hematopoietic cell types as well as HSPCs were predicted to receive Notch signals to some extent, raising the possibility of crosstalk and coordinated regulation between niche elements. It should be noted that, surprisingly, osteoblasts did not express high levels of lymphoid-associated factors (Figure 3A).

Using non-negative matrix factorization (NMF)³⁶, we identified higher-level communication modules in the bone marrow. We found that different non-hematopoietic subsets contributed to different modules, which we annotated based on the cell types involved (Figures 3G, S4F). The endosteal module, characterized by bone-specific gene families like NGF, ncWNT, and osteopontin, differed significantly from the MSC module which expressed families such as KIT, FGF, THY1. *CXCL12*^{hi} MSC subsets such as Adipo- and THY1+ MSCs contributed unique signals mainly to the MSC module whereas Osteo- or Fibro-

MSCs were involved in both the endosteal and MSC modules (Figure 3G). Furthermore, consistent with pathway-level CellChat analysis, VSMCs, and to a lesser extent, AECs, contributed to both endosteal and vascular signaling modules.

In sum, our analysis highlights the complex communication potential of human bone marrow mesenchymal and endothelial cells and suggests specialization of these cells in supporting different aspects of hematopoiesis. Particularly, Adipo- and THY1+ MSCs emerged as most transcriptionally specialized to support hematopoiesis through their high expression of key niche factors such as *CXCL12*, *KITLG*, *CSF1*, and *IL7*.

CODEX Multiplexed Imaging Reveals the Anatomy of the Human Bone Marrow Niche *In Situ*

We next sought to define the single-cell anatomy of the bone marrow cell types that we defined in our scRNA-Seq atlas. We employed whole-slide CODEX multiplexed imaging of 12 specimens, 8 of which were included in our transcriptomic atlas (Figure 4A). Guided by our scRNA-seq data, we designed and validated a 54-plex panel (53 antibodies + DAPI) (Figures 4A-B, Figure S5, Tables S4-S6, Data S1). For instance, in addition to canonical HSPC markers such as CD34 and CD38, we included markers derived from our scRNA-Seq data such as *SPINK2*, which was expressed most highly in primitive HSCs^{48,49} (Table S2). This increased the resolution to identify primitive HSPCs as the dynamic range of the CODEX assay is lower than flow cytometry where CD34 and CD38 alone are the gold-standard. Whole-slide CODEX images were acquired and subsequently segmented using Mesmer⁵⁰ to identify single cells. Cells were then annotated by unsupervised clustering, and in rare circumstances refinement by manual gating (STAR Methods). Importantly, after each round of clustering, cell labels were overlaid on the fluorescent image and clusters were iteratively visually inspected and corrected if necessary (Figures S6A, S5, Data S1). We computationally annotated 803,132 cells (91.6% of segmented objects) (Figures 4C, S5B) covering 32 cell types across 12 samples, including rare cells such as immunophenotypic HSCs (Lin- CD34+ CD38- CD45RA- CD90+) and Schwann cells (PLP1+ CD271+) (Figure 4C, Data S1, Table S4, Figure S6B). The overall cell type distribution was similar between individuals (Figure S6C, Table S1). The protein-level phenotyping also revealed certain interesting patterns, such as CD163 positivity of all observed macrophages and different clusters of GATA1+/- megakaryocytes (Data S1, Figure S6D).

We subcategorized the identified non-hematopoietic cells into AECs, SECs, VSMCs, endosteal cells, Adipo-MSCs, and THY1+ MSCs. Notably, despite some tissue processing artifacts, such as frequent loss of bone tissue, we were able to identify para-trabecular cells with non-hematopoietic expression profiles that we termed “endosteal”. Neither Osteo-MSCs nor Fibro-MSCs were initially detected computationally, suggesting that these cells were tied more closely to bone and are thus lost during tissue processing. Upon closer manual inspection, we indeed detected rare instances of these cells (Figure 4D), with Osteo-MSCs and osteoblasts being detected on the hematopoietic-bone interface, and Podoplanin+ Fibro-MSCs being detected largely in the middle of the bone region (Figures 4D-E, S6E). Additionally, 53% of the 422 manually annotated Osteo-MSCs overlapped with clustered “endosteal” cells, showing the endosteal population is enriched for osteolineage cells.

Furthermore, the protein expression profiles of the CODEX cell types mapped well to our scRNA-Seq data, with the matching cell types scoring very highly (Figures S6E-G). The final cell type annotations were used to visualize cellular topology in “cell phenotype maps” (CPMs, Figure 4F). Collectively, this annotated reference serves as a comprehensive map of the cellular topology of hematopoietic cells in the context of their microenvironment and demonstrates the complementarity of these two atlases to understand the cellular anatomy of human bone marrow.

Early Myeloid Progenitors and GMPs Localize to a Relatively Hyperoxygenated Arterio-Endosteal Niche

While multiple niches have been proposed to organize hematopoiesis^{2,5,34,35}, we sought to utilize an unbiased, statistically rigorous approach to answer the question of how hematopoietic and mesenchymal cells organize into cellular neighborhoods. Indeed, even with interactive visualization of the CODEX images, we noticed discrete patterns of cellular organization, such as peri-endosteal localization of MPO^{hi} early myeloid progenitors (EMPs), and erythroblastic islands composed of macrophages and erythroid precursors (Figure S7A).

We therefore performed unsupervised cellular neighborhood analysis as previously described⁵¹ and identified 15 cellular neighborhoods (CNs) that we manually annotated based on their enrichment of cell types using hypergeometric tests (Figure 5A). Given the challenge of bone detachment during the CODEX protocol, we first confirmed that neighborhoods enriched for “endosteal” cells were indeed close to manually annotated bone locations (STAR Methods, Figure S7B). We found expected neighborhoods such as the “Erythroid” neighborhood (CN13, CN15) consistent with erythroblastic islands, as well as novel neighborhoods such as one neighborhood of peri-arteriolar lymphoid cells (Figures 5A, S7C). We also found multiple mixed lineage neighborhoods (CN7, CN8) with all three hematopoietic lineages (Erythroid/Myeloid/Lymphoid). HSPCs, including more primitive SPINK2⁺ HSPCs, were found most in mixed-lineage neighborhoods (CN7, CN8), and SPINK2⁺ HSPC, to a lesser degree in Myeloid/Lymphoid (CN5) and Early Myeloid/Arteriolar (CN4) neighborhoods. Lymphoid progenitors such as CLP and immature B cells were found predominantly in mixed lineage (CN7,CN8), Plasma Cell/Arteriolar (CN3), and Peri-Arteriolar Lymphoid (CN2) neighborhoods. Two neighborhoods (CN4, CN6), enriched for GMP/myeloblasts and early myeloid progenitors, were strongly localized along the endosteal surface and around arterioles (Figure 5A-C). This was consistent with the endosteal signaling module that we identified using CellChat which had contribution from AECs and VSMCs (Figure 3G). We therefore termed them Early Myeloid/Arteriolar (CN4) and Early Myeloid/Endosteal (CN6), based on the enrichment of AECs and endosteal cells respectively. Notably, the Peri-Arteriolar Lymphoid neighborhood was not enriched for endosteal cells, unlike the Early Myeloid, Vascular/Endosteal/MSC, or Plasma Cell/Arteriolar neighborhoods (Figures 5A, S7B-C). Finally, the Vascular/Endosteal/MSC enriched neighborhood was highly enriched for Schwann cells, suggesting that this neighborhood is innervated. All of the arteriolar and mixed lineage neighborhoods (CN1-4, CN7-8) were enriched for Adipo- and THY1⁺ MSCs (Figure 5A). The neighborhoods were not overly biased towards one sample or by age (Figure S7D). Importantly, this

analysis also shows that the concept of the peri-arteriolar and endosteal niches holds up to unbiased statistical measurement of spatial association, and that in line with recent studies using mouse models⁵², adult human HSPCs do not preferentially occupy an endosteal or peri-arteriolar niche.

The importance of the endosteal niche has been reported in both healthy and malignant hematopoiesis^{5,53}. We therefore sought to further investigate interactions within this niche using both our CODEX and scRNA-Seq atlases. First, we found that arteriolar cells were near the trabecular bone more frequently than random expectation (Figures 5B, S7B), suggesting that the endosteal niche may actually combine with the arteriolar niche. Second, we found that HIF1 α levels were low in early myeloid progenitors and GMPs, suggesting that these cells are not experiencing hypoxia (Figure 5C), correlated with their localization near oxygen-supplying vessels. In contrast, more mature myeloid cells had high levels of HIF1 α (Figure S7E). Furthermore, early myeloid progenitors in non-arteriolar neighborhoods had higher levels of HIF1 α , suggesting that the spatial patterns of hypoxia are not due to intrinsic cell type differences (Figure 5D). Likewise, we found that early myeloid progenitors have the lowest hypoxia signature score in our scRNA-Seq atlas (Figure 5E), lending support to our CODEX findings. Given that the bone marrow is broadly a hypoxic environment, this finding suggests that a relatively oxygenated niche is associated with early myelopoiesis. Overall, our analysis suggests that the peri-arteriolar and endosteal niches are not, as is often discussed in the literature^{2,35}, two discrete niches and they seem to cooperate in some way important for early myelopoiesis.

We next aimed to integrate our findings based on our scRNA-Seq and CODEX data. First, we sought to validate specific expression patterns and interactions from our scRNA-Seq/CellChat predictions using an additional CODEX panel targeting some of the ligand-receptor pairs (Table S5). We found several instances of CODEX data matching the RNA/CellChat predictions. Specifically, we found that indeed CXCL12 expression was a key discriminating factor between arterial and sinusoidal endothelial cells (Figure S8A). We also found that Notch ligands JAG1 and DLL4 were expressed in both VSMCs and AECs. Finding DLL4 expression in VSMCs was surprising but has been previously reported^{54,55}. We also observed FGF2 expression in AECs and Angiopoietin-1 in MSC projections (Figures 3A, S8A, Table S3). Next, we aimed to interrogate whether CellChat-predicted interactions were recapitulated by spatial proximity in a dose-dependent fashion. We annotated cell types in the new CODEX images by reference mapping to our normal bone marrow (NBM) CODEX atlas and analyzed four predicted interactions involving mesenchymal and endothelial cells. We confirmed both well-established (VSMC-NOTCH3 and VSMC/AEC-JAG1, MSC-IL7 and CLP-IL7R) and previously underappreciated interactions (THY1+ MSC-COL1A1 and Monocyte-CD44, Erythroid/Megakaryocyte-Angiopoietin-1, and SEC-TIE2) (Figure S8B). We found that both strongly and less strongly CellChat predicted interactions could be recapitulated by spatial proximity and that there was a dose-dependent proximity of the target cell based on the level of receptor expression (Figures S8B-C).

While it is not possible to test every CellChat prediction, a useful resource could be one enabling prioritization of the CellChat predictions by the respective neighborhood

co-localization for a given cell type pair. To this end, we devised a unified CODEX/CellChat Effect Size (CCES) score for whether a given cell-type pair is both co-localized and has a high weight in the CellChat predicted signaling network (STAR Methods, Figures 5F, S8D, Table S8). Osteoblast-VSMC was one of the highest scoring interactions by this metric, which we found to co-localize and form an arterio-endosteal niche associated with lower hypoxia signature, demonstrating the utility of this metric in prioritizing important interactions (Table S8). Arterioles were found to be encased in Type 1 Collagen (Figure S8E), which has the highest expression in osteoblast cells (Figure 2B), suggesting one possible mechanism for the observed arterio-endosteal relationship.

Imaging-based Microanatomical Structural Analysis Reveals Peri-Adipocytic Localization of HSPCs

We next studied six types of key microenvironmental structures that are not well captured by conventional cell segmentation methods due to their complex shapes and/or sizes, including adipocytes, arterioles, bone, macrophages and their projections, sinusoids, and CXCL12+ stroma. We first employed manual annotation and thresholding to confidently annotate these structures (Figure S9A-B). Next, using point pattern analysis (STAR methods), we analyzed the distance between cell types or neighborhoods to each of these newly annotated structures⁵⁶. We used permutation tests to calculate the significance of identified proximity taking into account of the differential abundance of the identified structures (Figure S9C). We found that of all the measured structures, the arterioles were closest to bone (Figure 6A, $p=4.2E-6$). This supports the finding from neighborhood analysis that the endosteal niche was arteriole-rich (Figure 6B). In general, we found that the central niche, as defined by not being statistically closer to bone, was characterized by sinusoids, adipocytes, stroma, and macrophages, although adipocytes and sinusoids were further away from bone than stroma or macrophages (Figure 6A-B). Additionally, structural analysis confirmed that the previously mentioned erythroid niche corresponds to erythroblastic islands, as this niche was very proximal to macrophages, which were not captured in the neighborhood analysis due to their irregular shape and size (Figure S5D). The early myeloid neighborhoods were close to both arterioles and bone, providing further evidence of the arterio-endosteal EMP niche (Figure 6C).

Our results also validated our cell type annotations, as cell types co-localized with expected structures such as Adipo-MSCs-stroma ($p=3.8E-133$), macrophages-macrophages ($p=5.3E-156$), endosteal-bone ($p=6.8E-147$), and SEC-sinusoids ($p=9.8E-179$) (Figures 6D). Interestingly, we found that all MSCs were close to the arterio-endosteal niche, but Adipo-MSCs were found near sinusoids while THY1+ MSCs were not (Figure 6D). This suggests that Adipo-MSCs may be analogous to previously described HSC-supportive peri-sinusoidal MSCs^{40,57}.

We also discovered spatial patterns not apparent in our cell segmentation-based analysis. Both Lin- SPINK2+ and SPINK2- CD34+ HSPCs were found to be more proximal to adipocytes than expected by chance ($p=0.0198$ and $1.77E-6$ respectively, Figures 6D-E). Manual inspection of CODEX images revealed that both types of HSPCs were in frequent contact with adipocytes (Figure 6F, Table S7). HSPCs, but not more primitive SPINK2+

HSPCs, were also found to be close to macrophages ($p=5.23E-11$ and n.s. respectively, Figure 6E). Interestingly, GMP/Myeloblast cells as well as early myeloid progenitors were also close to adipocytes, although these cells were significantly closer to bone while HSPCs were not (Table S7). Mature myeloid cells, by contrast, were closest to sinusoids and not arterioles or bone, suggesting that maturation occurs in a central, peri-sinusoidal niche ($p=6.71E-68$, Figure 6D). Collectively, these data suggest spatial restriction of myeloid development where the earliest myeloid progenitors are specified in an adipocytic niche from HSPCs, migrate to a relatively hyperoxygenated arterio-endosteal early myeloid niche, and finally migrate to and mature near sinusoids that they may utilize to egress from the marrow and into the bloodstream (Figure 6G).

Unsupervised Reference Mapping Using Healthy Bone Marrow CODEX Atlas Reveals AML Stromal Expansion and Novel AML-MSC Enriched Neighborhoods

Understanding the spatial organization of the bone marrow microenvironment has significant implications for healthy hematopoiesis as well as disease states. For instance, acute myeloid leukemia (AML) has been described previously to interact with the endosteal niche to achieve chemoresistance⁵³. We applied our bone marrow CODEX panel to explore tumor evolution and microenvironmental changes using our healthy atlas as a reference. We profiled three diagnostic (Dx) and two paired post-therapy (PostTx) AML patient samples from iliac crest biopsies treated with venetoclax plus a hypomethylating agent (Ven/HMA) (Table S1). As controls, we also included three negative lymphoma staging bone marrow biopsies (NSM) from the iliac crest of 38-62 year-old patients. We selected AML patients with a detected *NPM1* mutation (*NPM1c* W288*fs) due to availability of a mutant specific antibody^{58,59} so that we could identify putative leukemic blasts. We employed RPCA reference mapping to classify cells in the AML and NSM samples to their closest counterparts in our healthy atlas (Figure 7A, STAR Methods). As expected, we found a significant increase in the fraction of myeloid cells, excluding mature myeloid, in diagnostic AML compared to NSM (47.6% versus 23.1% across samples; $p<2.2E-16$ by proportion test) (Figure 7B). This demonstrates the utility of our atlas to rapidly annotate hundreds of thousands of cells in both healthy and diseased contexts. Projecting these cell labels onto the segmentation masks revealed grossly appreciable architectural changes in diagnostic and post-therapy AML samples compared to NSM samples, such as a near total loss of adipocytes in the leukemic setting, and incomplete hematopoietic recovery 30 days post-therapy in the presence of residual blasts (Figure 7C). A more detailed look at the cellular landscape of AML also revealed marked stromal expansion, with 2-3 fold higher relative frequency of Adipo- and THY1+ MSCs in AML compared to NSM samples (Figure 7D). Next, we aimed to identify leukemic blasts using a classifier based on mutant *NPM1* and CD141 staining (Figures S10A-B, STAR Methods). Our classified blast percentages correlated with the *NPM1c* variant allele frequencies ($R=0.62$, Figure S10C). Using this approach, we identified leukemic populations including a GATA1+ *NPM1c*+ double positive population in the post-therapy sample of AML3 and individual *NPM1* mutant post-therapy cells (Figure 7E). We observed a ~2-fold increase in GATA1 expression in *NPM1* mutant blasts across both post-therapy samples, suggesting that these residual cells exhibited lineage plasticity (Figure 7F).

We next applied neighborhood analysis to AML Dx (n=3), PostTx (n=2), and NSM (n=3) samples separately and identified 15 cellular neighborhoods in each (Figures 7G, S10D). We first analyzed the composition of the negative staging marrow samples. This sample was nearly identical in neighborhood composition to the femoral head samples used to generate our atlas, including early myeloid/arterial/endosteal neighborhoods (NSM-Early Myeloid/Arterial, NSM-Early Myeloid/Endosteal, $R=0.83$ and 0.74 respectively). The correlation of the cell type enrichment profiles across neighborhoods was generally high, which confirmed the generalizability of the femoral head data to bone marrow specimens from iliac crest (median neighborhood $R=0.71$, Figures S10D-E). We also observed CD34+ cells contacting adipocytes in both adult and pediatric NSM samples (Figure S10F). Lastly, we observed all four MSC subsets in the NSM samples, suggesting that these MSC subsets exist across multiple anatomic sites (Figure S10G). In general, this analysis lends credence to the use of NSM samples as controls, as well as the generalizability of femoral head data.

In the diagnostic AML specimens, we found four blast-enriched neighborhoods – Dx-CN3/CN5/CN9/CN12 (Figure 7G, S11A), two of which were enriched for MSC populations. Notably, THY1+ MSCs were significantly enriched exclusively in these blast-enriched neighborhoods (Figure 7G). No neighborhoods identified were exclusive to just one sample (Figure S11B). We observed neighborhoods similar to healthy samples as well, such as a plasma cell-AEC neighborhood (Dx-CN8) and lymphoid neighborhoods (Dx-CN6/CN14/CN15) (Figures 5A, 7G). Dx-CN9, a blast-enriched neighborhood, was also enriched for AECs, endosteal cells, and Adipo- and THY1+ MSCs, very similar to the healthy early myeloid progenitor niche. Interestingly, we found that both post-therapy blast-enriched neighborhoods (PostTx-CN4/CN14) resembled Dx-CN9 and NSM-Early Myeloid/Arterial/Endosteal neighborhoods, the arterio-endosteal early myeloid niche (Figure S11C). Notably, PostTx-CN4 and Dx-CN9/CN12, but not PostTx-CN14, were statistically closer to bone as determined by point pattern analysis (Figure S11D). We also observed that recovering Post-Tx GMP/early myeloid progenitor neighborhoods did not contain AECs/Endosteal cells (PostTx-CN8) which could reflect a difference between steady-state and emergency myelopoiesis, as previously reported⁶⁰ (Figure 7G). In general, this data hints at conserved spatial patterns marking hematopoietic recovery.

Lastly, we examined the intrinsic properties of *NPM1* mutant blasts. Both diagnostic and post-therapy *NPM1* mutant blasts had low HIF1 α levels consistent with GMPs and early myeloid cells (Figure S11E). Comparing paired samples, PostTx cells had slightly decreased BCL2 and increased mitochondrial Complex IV expression after therapy ($p < 2.2E-16$ for both, Figures S11F-G). This could reflect decreased reliance of these PostTx cells on the BCL2 pathway and compensatory increase in mitochondrial mass to circumvent venetoclax-mediated suppression of BCL2 and oxidative phosphorylation^{61,62}. These data provide *in vivo* evidence of AML-MSC co-localization and suggest that the arterio-endosteal early myeloid neighborhood that we identified may play a role in therapy resistance, although more samples will need to be studied to draw more general conclusions.

Discussion

We report the most comprehensive healthy human bone marrow reference to date, covering both hematopoietic and non-hematopoietic cell types and their spatial organization. We found that the bone marrow microenvironment contains at least six mesenchymal subsets, two endothelial cell subsets, and one vascular smooth muscle population. Our data clearly demonstrate the limitations of bone marrow aspirates as a sample source, where Adipo-MSCs are heavily favored and subsets like Fibro-MSCs are not captured at all^{16-18,63}. We identified markers to sort MSC subpopulations and pinpointed the rare Podoplanin+ Fibro-MSCs as the closest to the ISCT-defined mesenchymal stem cell. By contrast, Adipo- and THY1+ MSCs were found to be the most interactive with hematopoietic cells and not part of the endosteal signaling module. Adipo-, THY1+, Osteo-, Fibro-, and APOD+ MSCs also had significantly different profiles of support factor expression. These findings collectively demonstrate the utility of our atlas for understanding differential functions of MSC subsets.

We elucidated the spatial organization of the human bone marrow on a systematic, single-cell level. While recent studies have assessed human bone marrow organization using 7-color immunofluorescence⁶⁴, our work more deeply dissected the cellular differentiation continua of both hematopoietic and non-hematopoietic lineages by virtue of our 53-antibody panel. Our healthy atlas captures the spatial relationships of 32 cell types with respect to each other and to many manually annotated, difficult to segment cell types/structures. This atlas links mesenchymal heterogeneity on the transcriptional level to discrete spatial phenotypes *in situ*, for instance in the case of more central Adipo/THY1+ vs. endosteal Osteo/Fibro-MSCs. Our spatial findings were generalizable across anatomic sites as well, as our observations in femoral head bone marrow were closely recapitulated in negative lymphoma staging marrows from iliac crest. Our data will serve as a powerful tool for uncovering bone marrow niche biology in health and disease by enabling rapid reference mapping and label transfer to study how these otherwise conserved spatial patterns become dysregulated or co-opted, as we show in the AML vignette.

Our findings also highlight the importance of measuring all bone marrow cell types and microanatomical structures simultaneously. For instance, we extend prior findings⁶⁵ of an HSPC-adipocyte spatial relationship by showing that the relationship is strongest for more primitive^{48,49} SPINK2+ HSPCs and is stronger than any spatial association between SPINK2+ HSPCs and other structures. This finding is of particular importance as adipocytic density increases as humans age and local adipocytic density is markedly reduced in the context of acute leukemia. Bone marrow adipogenesis has been proposed as a negative regulator of hematopoiesis in mice⁶⁶, but a growing body of evidence has proposed adipocytes as both stemness-maintaining and supportive of stress hematopoiesis⁶⁷⁻⁷⁰, and so whether this interaction represents a hematopoiesis-supportive or suppressive interaction in humans warrants further investigation. We surprisingly did not find an endosteal or perivascular preference for human HSPCs in femoral head or iliac crest. This data does not mean that sinusoid/osteoblast-HSC communication is unimportant, only that human HSPCs are not closer to sinusoids than expected by chance in older adult samples. Similar

profiling efforts across the spectrum of aging will help reveal the relative contribution of these structures at different developmental stages.

Our data also have important implications for use and interpretation of mouse models. While mice and humans share mesenchymal subsets, humans have unique MSC populations like THY1+ and APOD+ that warrant further study. Fibro-MSCs in humans closely resemble early mesenchymal progenitors in mice, but Podoplanin has not been reported as a key early MSC marker in mice. Spatially, we found significant divergence from mouse biology. For instance, GMPs have been reported to be randomly scattered in the bone marrow⁶⁰. To our knowledge, the arterio-endosteal early myeloid niche has not been reported in mice. Perhaps most importantly, mouse bone marrow adiposity is several orders of magnitude below its human counterpart⁷¹, suggesting that murine samples may be representative of a much younger bone marrow microenvironment even in aged mice and therefore could explain why the HSPC-adipocyte spatial relationship is not observed in mice⁷².

We envision that this resource will have value to contextualize diseased bone marrow, as illustrated in our AML case study. While the small sample size and specific use of one treatment approach in this demonstrative vignette precludes broad generalization of our findings, we show that it is both feasible and informative to use CODEX to map blast-stromal neighborhoods to study the bone marrow microenvironment. The preliminary findings that Adipo- and THY1+ MSCs were expanded in AML samples and enriched in leukemic neighborhoods provide further rationale to study blast-stromal interactions in AML. Furthermore, our ability to detect and map leukemic minimal residual disease and rare post-therapy cells showed an erythroid GATA1+ enrichment in resistant blasts, highlighting the translational potential of bone marrow CODEX to detect and find prognostic spatial biomarkers of residual disease.

Limitations of the Study

Our study cohort comprises a representative sample of 12 individuals aged 52-80 years, approximating a relatively “old” microenvironment. Therefore, significant future work remains to determine the temporo-spatial landscape of hematopoiesis in humans. Additionally, adipocytes are a major component of the bone marrow microenvironment whose potential importance was highlighted in our spatial analysis. However, due to their size and fragility, we were not able to capture them in our scRNA-Seq atlas. Future efforts to sequence bone marrow adipose tissue would help resolve the mechanisms underlying HSPC-adipocyte co-localization.

Image analysis in the context of mapping complex cellular topologies, so-called “neighborhoods,” is in its infancy. Tools do not currently exist to consider both higher-order microenvironmental structures (e.g., arterioles or lymphoid follicles) and cell types for collective neighborhood analysis. Furthermore, 2D imaging is limited to distances that can be measured by observable analytes within a 2D plane within a complex 3D structure. This is particularly problematic for very rare cell types like hematopoietic stem cells, which could have supportive structures directly above or below the captured plane but would not have a spatial association captured. As 3D multiplexed imaging and corresponding analytical

approaches develop, we will gain a more comprehensive quantitative view of the human bone marrow.

STAR Methods

Resource Availability

Lead Contact—Further information and requests for reagents and resources should be directed to and will be fulfilled by the lead contact, Kai Tan (tank1@chop.edu).

Materials Availability—All materials used in this study are commercially available, as specified in the Key Resources Table.

Data and Code Availability—Software and source code created for this study are publicly available and can be obtained at GitHub at <https://github.com/tanlabcode/spatial-bonemarrow-atlas>. Single-cell RNA-Seq data can be downloaded from NCBI Gene Expression Omnibus (GEO) database with the accession number GSE253355 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE253355>, code: kloreiqqrnqfvaz). This includes the raw sequencing reads, count matrices, and annotated Seurat object.

Primary imaging data and processed CODEX data are publicly available and can be downloaded from FigShare at:

- Processed CODEX Data (Seurat Objects): <https://figshare.com/s/deca32e656983ce0b038>
- Normal Bone Marrow CODEX Images: <https://figshare.com/s/e4e20cea61151a89d582>
- Normal Bone Marrow CODEX Segmentation Masks: <https://figshare.com/s/b964d446e2860d3f8af6>
- Normal Bone Marrow H&E Images: <https://figshare.com/s/c57ace155970e21cdfdd>
- AML and NSM CODEX Images: <https://figshare.com/s/1b3d22ae14f4a448f21c>
- AML and NSM CODEX Segmentation Masks: <https://figshare.com/s/6951f96a907e615ebe09>
- AML and NSM H&E Images: <https://figshare.com/s/ceb6f2928e1c01765331>

This includes full multiplexed images for the normal bone marrow atlas, AML, NSM samples, corresponding cell segmentation masks and H&E images as well annotated Seurat objects which contain unnormalized and normalized marker expression values for each cell.

Both scRNA-Seq and CODEX data are publicly available to explore interactively through Vitessce widgets hosted at <https://cscb.research.chop.edu/index.php/bm-data>. The username is “demo” and the password is “BMatlas2023” (case sensitive, without quotes).

Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Experimental Model and Study Participant Details

Human clinical sample acquisition—Bone marrow samples were obtained from patients undergoing total hip arthroplasty surgery. We reasoned that osteoarthritis patients undergoing total hip arthroplasty surgeries, where the femoral bone is removed and discarded as surgical waste, would be a good source of hematologically healthy bone marrow. We considered that these samples were representative of normal hematopoiesis due to two major factors: first, studies have reported that 75-90% of individuals over 75 years of age will develop some degree of osteoarthritis⁷³, making the condition within the range of aging-associated changes; and second, osteoarthritis is a peri-articular disease not associated with hematologic complications⁷⁴. We only sampled at the distal site to the articular cartilage and only collected morphologically normal trabecular bone and associated bone marrow. For femoral heads profiled via scRNA-Seq or CODEX, the age range was 52-80 years, with a median of 65.5 (IQR: 58.5-71.25) years. Patients filled out a questionnaire explaining whether they had certain common medical conditions which are tabulated in the clinical metadata table. All femoral head patient samples were deemed as medical waste and IRB exempt by the University of Pennsylvania Institutional Review Board. Only non-identifiable patient information is collected for each sample by the clinical staff. Researchers were never given access to medical records. Acute myeloid leukemia (AML) and negative staging marrow (NSM) samples were obtained from the Hospital of the University of Pennsylvania's Pathology sample repository. These formalin fixed paraffin embedded (FFPE) samples are collected via trephine bone marrow biopsy from iliac crest, fixed in 10% neutral buffered formalin (NBF) for a minimum of 1 hour, decalcified with Formical for 2 hours, and rinsed with water for 10 minutes before paraffin embedding. All AML and NSM samples were appropriately collected under University of Pennsylvania IRB #826755. All patient clinical information is shown in Table S1.

Method Details

Femoral head cell isolation—Femoral heads obtained from total hip arthroplasty surgeries were cut coronally in 4-6 mm thick slices by the Anatomical Pathology Lab at Penn Presbyterian Medical Center. Tissue samples were transported on ice in AlphaMEM (Gibco) with 3% FBS (HyClone), 1% penicillin-streptomycin (Gibco), 0.02 mM L-ascorbic acid (Sigma) and 0.3 mM L-Glutamine (Gibco). Samples were placed in a petri dish and cut with surgical wire cutters into approximately 1 mm cubed pieces, weighed and transferred to a 15 mL centrifuge tube. For CODEX, a dental biopsy drill was used to harvest cylindrical samples, cut to about 4 mm thick, and fixed in 4% paraformaldehyde (PFA) (EMS). For scRNA-Seq and cell sorting, 1 mm of minced tissue was digested in filtered 4 mg/mL dispase II (Sigma), and 2 mg/mL collagenase I (Worthington Biochem) in PBS for one hour on a rocker at 37°C. The cell suspension was then filtered through a 100 µm cell strainer (Fisher) and remaining tissue was washed 5 times in culture medium (AlphaMEM with 15% FBS, 1% P/S, 0.1 mM LAA, 1 mM L-glutamine) and strained. Total collected suspension was pelleted at 300 RCF for 10 minutes and resuspended in MesenCult Proliferation medium (STEMCELL Technologies, Cat# 05411), and immediately taken for immunomagnetic population isolation (see “Single-cell RNA sequencing” and “Mesenchymal cell fluorescence assisted cell sorting” sections).

Micro-CT BV/TV determination—Bone tissue cores were scanned with a micro-computed tomography scanner (Model MicroCT35, Scanco Medical AG, Switzerland) at a 6- μ m isotropic voxel size. All images were analyzed with the Scano scanner software and a region of 4200 μ m in diameter and 900 μ m in height was selected and smoothed with a Gaussian filter (sigma = 1.2, support = 2.0) prior to binarization with a minimum threshold of 365 and bone volume to total volume (BV/TV) calculated.

Single-cell RNA sequencing—Digested tissue samples were subjected to immunomagnetic RBC depletion reagent (STEMCELL Technologies, Cat# 18170). 20 μ L of RBC depleted cells were put on ice, while 67% of the remaining cells were used for immunomagnetic CD45 depletion using EasySep Human CD45 Depletion Kit II (STEMCELL Technologies, Cat# 17898) to isolate non-hematopoietic cells in an unbiased fashion, while 33% of cells were subjected to immunomagnetic CD34 enrichment using EasySep Human CD34 Positive Selection Kit II (STEMCELL Technologies, Cat# 17856). Following enrichment, each of the three cell fractions (RBC depleted, CD45 depleted, CD34 enriched) were resuspended at a concentration of 1000 cells/ μ L in PBS + 0.04% BSA. Cells were pooled and loaded into the 10x Genomics Chip G (Cat# 1000127). We used Chromium Next GEM Single Cell 3' GEM, Library & Gel Bead Kit v3.1 (10x Genomics, Cat# 1000121) and followed the protocol as provided by 10x Genomics. Sequencing was performed at the High Throughput Sequencing Core at CHOP using a NovaSeq 6000 system. Sequencing parameters were set according to 10x Genomics recommendations.

Bone marrow decalcification—Femoral head samples were fixed for 24 hours in 4% paraformaldehyde (Electron Microscopy Sciences, Cat# 15710-S diluted to 4%) in PBS (Gibco, Cat# 14190-136) at 4°C. Samples were washed 3 times for 30 minutes in PBS at 4°C. Samples were decalcified in 0.5M EDTA at pH8 (Invitrogen, Cat# 15575-038) for 8 days. The EDTA was changed every 2 days. After decalcification, samples were washed 5 times for 1 hour in PBS and fixed for 24 hours in 4% paraformaldehyde at 4°C. Samples were processed and embedded in paraffin by the Children's Hospital of Philadelphia's pathology core. The AML and NSM samples were obtained from the University of Pennsylvania Department of Pathology sample repository. Bone marrow biopsies were fixed in buffered zinc formalin for a minimum of 1 hour. Bone marrow biopsies were then decalcified in Formical-4 (StatLab, SKU# 1214-1) for 2 hours. Samples were rinsed with running water for a minimum of 10 minutes. Samples were placed in neutral buffered formalin and submitted for paraffin embedding.

Antibody panel design and validation—Antibodies for CODEX were chosen based on a combination of literature (esp. (Kathryn Foucar's Bone Marrow Pathology 4th edition textbook⁷⁵, reference⁷⁶, STEMCELL Technologies' recommendations <https://www.stemcell.com/human-hematopoietic-stem-and-progenitor-cell-phenotyping-panels.html>) and our scRNA-Seq data. Mesenchymal markers were entirely derived from our scRNA-Seq data and sorted assays. Antibodies were chosen for testing which had FFPE validation by the vendors and high signal-to-noise ratio on validation. Antibodies were further validated by co-labeling and cell-type appropriate expression, which is shown in the Supplemental Cell Type Annotation Guide (Data S1).

CODEX antibody conjugation—Akoya antibodies were purchased pre-conjugated to their respective CODEX barcodes (Table S5). All other antibodies were custom conjugated to their respective CODEX barcodes (Table S5) according to Akoya's PhenoCycler-Fusion user guide using the antibody conjugation kit (Akoya, Cat# 7000009). Briefly, 50 µg of carrier-free antibodies were concentrated by centrifugation in 50kDa MWCO filters (EMD Millipore, Cat# UFC505096) and incubated in the antibody disulfide reduction master mix for 30 minutes. After buffer exchange of the antibodies to conjugation solution by centrifugation, addition of conjugation solution and centrifugation, respective CODEX barcodes resuspended in conjugation solution were added to the concentrated antibody and incubated for 2 hours at room temperature. Conjugated antibodies were purified by 3 buffer exchanges with purification solution. 100 µl of antibody storage buffer was added to the concentrated purified antibodies.

CODEX staining—CODEX staining was done using the sample kit for PhenoCycler-Fusion (Akoya, Cat# 7000017) according to Akoya's PhenoCycler-Fusion user guide with modifications to include a photobleaching step and overnight incubation in antibodies at 4°C. FFPE samples were sectioned at 5 µm thickness and mounted onto charged slides (Leica, Cat# 3800080). Sample slides were baked for 3 hours at 65°C. Sample slides were deparaffinized in Histochoice clearing agent (VWR, Cat# H103-4L) and rehydrated in a graded series of ethanol (2 times 100%, 90%, 70%, 50%, 30% and 2 times ddH₂O). Antigen retrieval was performed in 1x citrate buffer pH6 (Sigma, Cat# C9999 diluted to 1x in ddH₂O) with a pressure cooker for 20 minutes. After equilibrating to room temperature, sample slides were washed 2 times with ddH₂O and submerged in a petri dish containing 4.5% H₂O₂ and 20 mM NaOH in PBS (bleaching solution) for photobleaching. The petri dish was sandwiched between two broad-spectrum LED light sources for 45 minutes at 4°C. After 45 minutes, sample slides were transferred to a new petri dish with fresh bleaching solution and photobleached for another 45 minutes at 4°C. Sample slides were washed 3 times in PBS and then 2 times in hydration buffer. Sample slides were equilibrated in staining buffer for 30 minutes and incubated in the antibodies diluted in staining buffer plus N Blocker, G Blocker, J Blocker, and S Blocker overnight at 4°C. After antibody incubation, sample slides were washed 2 times in Staining Buffer and fixed for 10 minutes in 1.6% paraformaldehyde in storage buffer. Sample slides were washed 3 times in PBS and incubated in ice cold methanol for 5 minutes. After incubation in methanol, sample slides were washed 3 times in PBS and incubated in final fixative solution for 20 minutes. The sample slides were then washed 3 times in PBS and stored in storage buffer for up to one week prior to imaging.

CODEX imaging—CODEX reporters were prepared according to Akoya's PhenoCycler-Fusion user guide and added to a 96-well plate. The PhenoCycler-Fusion experimental template was set up for a CODEX Run using Akoya's PhenoCycler Experiment Designer software according to Akoya's PhenoCycler-Fusion user guide. Details on the order of fluorescent CODEX barcodes and microscope exposure times, can be found on (Table S6). The PhenoCycler-Fusion experimental run was performed using Akoya's Fusion 1.0.6 software according to Akoya's PhenoImager Fusion User Guide. Images were taken on

the PhenoImager Fusion microscope using default settings and pre-processed (stitching, registration, background subtraction) by the Fusion software.

H&E staining and registration—Slides were removed from the microscope after CODEX imaging with the PhenoCycler-Fusion platform and kept in Histochoice for 3-5 days. The flow cell was gently removed and then the tissue slide was sent to CHOP's pathology core for standard hematoxylin and eosin (H&E) staining. Imaging was performed on the PhenoCycler Fusion microscope using the brightfield mode. 40x H&Es were registered to the 20x CODEX imaging space using wsireg (<https://github.com/NHPatterson/wsireg>) with affine and rigid transformations, the default parameters, with the DAPI and NaK-ATPase channels as input from the CODEX image to label the nucleus and membrane (to match with hematoxylin and eosin) respectively.

Mesenchymal cell fluorescence assisted cell sorting—Femoral head samples from donors were digested as described above and subjected to RBC depletion and CD45 depletion using MACS (STEMCELL Technologies, Cat# 18170 and 17898). Antibodies for FACS were purchased from BioLegend unless otherwise specified. Anti-LEPR was purchased from R&D systems (Clone 52263, Cat# FAB867P). Isolated cells were stained for 15-30 minutes using anti-CD90-Bv650 (Clone 5E10), anti-CD38-Bv785 (Clone HIT2), anti-CD45- FITC (Clone HI30), anti-LEPR-PE (R&D Systems, Clone 52263), anti-Podoplanin-PE/Dazzle (Clone NC-08), anti-CD56-PE/Cy7 (Clone 5.1H11), anti-VE-Cadherin-Alexa Fluor 647 (Clone BV9), and anti-CD235ab-APC/Fire (Clone HIR2). Cells were spun down and resuspended in PBS + 2% FBS containing DAPI to exclude dead cells. Fluorescence minus one controls were used to confirm the gating strategy, and all mesenchymal cells were gated as CD45⁻ DAPI⁻ CD38^{lo} CD235ab⁻, hereby referred to as hem-negative. Fibro-MSCs were defined as hem-negative PDPN⁺ VE-Cadherin⁻. Osteolineage cells were defined as heme-negative PDPN⁻ VE-Cadherin⁻ CD56⁺. THY1⁺ MSC were defined as hem-negative PDPN⁻ VE-Cadherin⁻ CD56⁻ LEPR⁺ CD90⁺ and Adipo-MSCs were defined as PDPN⁻ VE-Cadherin⁻ CD56⁻ LEPR⁺ CD90⁻. Cells were directly sorted into MesenCult Proliferation Kit (STEMCELL Technologies, Cat# 05411) and growth media containing 25mM HEPES and cultured or used directly for downstream experiments.

Proliferation/Cell culturing—Sorted cells were seeded in MesenCult Proliferation medium (STEMCELL Technologies, Cat# 05411) for passaging. The population doubling was calculated as $3.32 \cdot \log((\text{cell count at passage})/(\text{cells seeded}))$. The cumulative population doubling is reported. The cumulative population doubling for the four cell types over eight passages was analyzed by a 2-way ANOVA followed by a Bonferroni post-hoc test comparing each cell type at each passage to Fibro-MSCs.

CFU-F assays—Cells were seeded immediately after fluorescence assisted cell sorting at 145 cells/cm² in a 6-well or 12-well cell culture treated dish and maintained in MesenCult Proliferation Kit (STEMCELL Technologies, Cat# 05411), until colonies could be visualized under the microscope (8-12 days after beginning the assay). Cells were washed in PBS and stained with crystal violet (3%). Colonies were identified by morphology and manually counted as clusters with greater than 30 cells forming a tight grouping.

Differentiation potential assessment—Sorted Fibro-MSCs were expanded in culture before plating for differentiation assays to achieve the necessary input cell number per patient. At 90% confluence, growth medium was switched to osteogenic induction medium (AlphaMEM with 10% FBS, 1% P/S, 0.1 mM LAA, 1 mM L-glutamine, 10^{-2} μ M DMSO, and 1.8 mM KH_2PO_4) or adipogenic induction medium (AlphaMEM with 10% FBS, 1% P/S, 0.1 mM LAA, 1 mM L-glutamine, 0.5 mM isobutylmethylxanthin, 60 μ M indomethacin, 0.5 μ M hydrocortisone, and 10 μ g/mL insulin). Cells were harvested at day 14 for qRT-PCR analysis and at day 28 for Alizarin Red (osteogenic), Oil Red O (adipogenic), or Alcian Blue (chondrogenic) staining. For qPCR analysis (Table S9), total RNA was isolated using a phenol-chloroform extraction. We were unable to extract RNA from chondrogenic pellets likely due to the 3D structure, and so focused on qPCR for adipogenic and osteogenic differentiations, relying on the Alcian blue staining alone for chondrogenic. 1 μ g of RNA was used for reverse transcription (High-Capacity cDNA Reverse Transcription Kit, Applied Biosystem). qPCR was performed using Power SYBRTM Green PCR Master Mix. Samples were normalized to endogenous control *GAPDH* and fold change from the median control Ct value calculated using the ddCT method⁷⁷.

Mesenchymal and hematopoietic cell co-culture assay—One day prior to hematopoietic progenitor cell isolation, sorted MSC cell subsets (Fibro-, THY1+, Osteo-, and Adipo-MSCs) were seeded at 10,000 cells/cm² on a 6-well dish (Falcon, 353046). Hematopoietic progenitor cells were isolated from digested femoral head cell suspensions by performing RBC depletion (STEMCELL Technologies, Cat# 18170) followed by CD34+ cell selection (STEMCELL Technologies, Cat# 17856) with the StemCell EasySep system. The MSC cell subsets were washed twice with PBS and growth medium changed to DMEM with 15% FBS, 0.1 mM LAA and 1% penicillin-streptomycin. CD34+ isolated cells were added to each well at 10,000 cells/cm² and an additional well seeded with only CD34+ isolated cells. After three days, 1 mL medium was added. On day six, cells in suspension were combined with trypsinized adherent cells, pelleted, and resuspended in PBS with 2% FBS. Cells were blocked with Fc receptor blocker (Biolegend, Cat# 422302) and then stained with Biolegend antibodies, unless otherwise stated, CD61-BUV395 (BD Biosciences, Cat# 745717), CD19-BV510 (Cat# 302241), CD14-BV605 (Cat# 367125), CD33-BV650 (Cat# 303429), CD38-BV785 (Cat#, 303530), CD15-FITC (Cat# 394705), CD235ab-PerCP/Cy5.5 (Cat# 306613), CD66a/c/e-PE (Cat# 342303), CD45-APC/Cy7 (Cat# 368516), CD163-AF647 (Cat# 333619), CD34-PE/Cy7 (Cat# 343615) and CD71-AF700 (Cat# 334130). DAPI was used as a live/dead stain. Flow cytometry was performed on an Aurora spectral analyzer with 5 lasers and 64 detectors (Cytek Biosciences) through the Children's Hospital of Philadelphia Flow Cytometry Core. Spectral unmixing and compensation was accomplished with single stained compensation beads (Invitrogen). Differentiation potential was assessed as described previously⁷⁸ – that is, gating of lineage uncommitted CD34+ CD38- cells by excluding cells expressing markers of erythroid, B-cell, myeloid, and monocytic differentiation.

Figure making and plotting—Nearly all graphical figures were generated using R v4.0.5 and subsequently edited in Adobe Illustrator. Napari (doi:10.5281/zenodo.3555620) was used for generation of CODEX image figures, including fluorescent CODEX images, masks,

and H&E images. Images in the manuscript figures are contrast-adjusted for illustrative purposes. For images, in certain cases Adobe Photoshop/Illustrator was used to globally adjust brightness, contrast, or threshold for visualization purposes. GraphPad Prism v9 was used to plot Figure S3D and perform statistical testing in Figure 2F.

Quantification and Statistical Analysis

scRNA-Seq data processing, annotation, and pathway analysis—scRNA-seq data was demultiplexed, aligned, and count matrices generated using Cell Ranger v6.1.2 aligning to GRCh38 with intronic regions included. Cell Ranger matrices were directly imported into Seurat v4. Cells with 300-10,000 unique genes expressed, less than 10,000 unique UMIs, and less than 10% mitochondrial reads were retained in the final dataset. Individual samples were processed by log-normalization, the top 2,000 features were identified, the data was scaled, a principal component analysis (PCA) was computed, and the data was clustered via Louvain clustering using the top 30 principal components (PCs). We used DoubletFinder⁷⁹ to remove doublets on a per-sample basis using the top 30 PCs, an expected doublet rate equal to the number of cells in the sample/125000, and other parameters remained as default. Doublet cells were removed, and samples were combined without computational integration. The full dataset was processed as above and visualized using UMAP with the top 30 PCs. Differentially expressed genes in each cluster were identified using the FindAllMarkers function in Seurat v4 with the Wilcoxon Rank Sum downsampled to maximum 1,000 cells per cluster and these results are recorded in Supplemental Table S2. Cells were annotated by marker gene expression, and likely artifactual cells (only found in one sample or likely residual doublets) were removed from analysis using VisCello for interactive visualization and artifact selection and removal⁸⁰, followed by dimensionality reduction and clustering in an iterative process to refine the dataset. Mesenchymal and endothelial cell clusters were computationally isolated for further analysis. Subset analysis was conducted similarly as above, first calculating the top 2,000 variable features, then scaling, recomputing a PCA, Louvain clustering using the top, and visualization using UMAP with the top 50 PCs. Likely artifactual cells were removed using VisCello and these subsets were iteratively re-processed and visualized. AUCell⁸¹ was used for pathway analysis with the GSEA human Hallmark gene set (<https://www.gsea-msigdb.org/gsea/msigdb/human/collections.jsp>)⁸². One population, RNA^{lo} MSCs, were excluded from downstream analysis because of their indeterminate QC profile which was consistent with highly differentiated cells like neutrophils or RBCs but did not match a known cell type and thus could also be artifact. Differentiation states of MSCs were inferred using CytoTRACE (v0.3.3) with default parameters using the log-normalized counts. Further details are provided in the accompanying source code.

External dataset mapping—External datasets were downloaded per the instructions on the respective manuscripts. Samples from De Jong *et al.* were downloaded from EBI (E-MTAB-9139), and myeloma samples were excluded (CBM1_1, CBM1_2, CBM2_1, CBM2_2, S9, S10, S11, S12, and S13). Count matrices were generated using Cell Ranger 4.0.0 with default parameters. Samples were integrated and CXCL12+ clusters were exported. Data from Li *et al.* were downloaded from NCBI GEO (GSE190965) and directly read into Seurat and reprocessed as in our dataset described above. For Wang *et al.* data,

we downloaded feature barcode matrices from NCBI GSE147287 and read into Seurat after concatenating the spliced and unspliced matrix files and reprocessed as above. We downloaded the Ennis *et al.* data from GSE227903 and used the SeuratDisk package to convert from the provided aggregated h5ad data to a Seurat object and reprocessed the data. For Triana *et al.* data, we retrieved the Seurat object directly from Figshare (https://figshare.com/articles/dataset/Expression_of_97_surface_markers_and_462_mRNAs_in_70017_cells_from_healthy_young_healthy_old_and_leukemic_human_bone_marrow/1339765 1). When reprocessing data for was required for any dataset, we retained cells with 100-10,000 unique genes expressed, and less than 10% mitochondrial reads were retained. We identified the top 2,000 variable features, and recomputed a PCA before performing reference mapping.

For samples in public data in which annotations were available, mesenchymal cells were extracted and reference mapped to our MSC reference using reciprocal PCA (RPCA) in Seurat, using the FindTransferAnchors function with the top 50 dimensions followed by the MapQuery function to project the data onto our reference atlas. Default parameters were used generally, except for the Triana *et al.* (k.weight=30) and Ennis *et al.* data (k.filter=10, k.score=17, k.weight=30) where the very small cell numbers necessitated lowering the k parameters. In cases where annotations were not available, we mapped to our whole reference (hematopoietic + non-hematopoietic cells) and showed just the results for cells mapping to non-hematopoietic cells. The exception to this is the De Jong *et al.* data, which we filtered down to mesenchymal cells using marker gene expression (*VIM*, *PTPRC*, *CXCL12*, *CD38*) ourselves due to the size of the unfiltered dataset. More details for this analysis including preprocessing can be found in the accompanying source code.

Inflammation signature analysis—GSEA Hallmark Inflammatory Response scores were computed using AUCell⁸¹ 300 cells per cell type in Figure S2A. To compare inflammatory cytokine expression across datasets, we used Chen *et al.* data⁸³ as a representative pediatric atlas, the Human Cell Atlas Immune Cell Atlas Bone Marrow (“hcbam40k” installed using Seurat), and Li *et al.* data¹⁷. We filtered our dataset to just MSCs and re-calculated the GSEA Hallmark Inflammatory Response signature using AUCell with 5000 randomly chosen cells per dataset. We also included publicly available scRNA-Seq data from cultured primary iliac-crest-aspirate-derived human MSCs treated with and without 15 ng/mL TNF α and 10 ng/mL IFN γ for 24 hours⁸⁴ (GSE115149). We called the TNF α /IFN γ treated MSC “Inflamed_MSC” and the untreated MSCs as “Control_MSC”. We also included Li *et al.* MSCs since they are representative bone marrow aspirate (and therefore can be considered “normal”) in this analysis. Li *et al.* MSCs were isolated as described in the External Dataset Mapping section. Statistical testing was performed using two-sided Welch’s two sample t-test with each population tested against the inflamed MSCs.

Cellular interaction analysis—CellChat was used for cellular interaction analysis using the CellChatDB database. As described in the original manuscript³⁶, scRNA-Seq values were projected onto a protein-protein interaction network to account for typical dropout issues. The significant cell-cell interactions were determined using CellChat default

parameters, with population size accounted for and “trimean” used for calculating mean gene expression. Signaling family (‘Pathways’ in the original manuscript) annotations were taken from the original manuscript and represent groups of signaling molecules that function in related ways, and are often in the same protein family. For the Non-negative Matrix Factorization (NMF) signaling pattern analysis, the value of K was selected when the Cophenetic and Silhouette scores began to drop. Our analysis was guided by (<https://htmlpreview.github.io/?https://github.com/sqjin/CellChat/blob/master/tutorial/CellChat-vignette.html>) and further details can be found in our source code.

CODEX data processing—Whole-cell segmentation was performed using Mesmer⁸⁵ for each image. To generate the necessary input of a two-channel TIFF, we used DAPI for the nuclear channel and a composite channel of CD45, Vimentin, and NaK-ATPase for the membrane channel generated through manual rescaling of each individual channel followed by merging the channels (see source code for further details). Mean pixel intensity was extracted from each cell segmentation mask, yielding a cell x protein matrix which was carried forward for analysis in Seurat v4. Cells with very low or high raw DAPI expression (<10 or >250 on a UINT8 scale) were removed. Images were cropped if there were instances of tissue detachment during the run, or if areas had noticeable artifactual staining. All such cropping is carefully documented in the source code. Finally, we computed the sum of all the marker expression in each cell and removed cells falling above the 99th percentile of total expression or below the 1st percentile of total expression to reduce the impact of reporter precipitates and similar artifacts. One Seurat object was created for each sample using this method. Each sample was then internally normalized using Seurat’s implementation of centered log-ratio (CLR) for each feature independently (margin=1) which is recommended for analysis of protein expression as in CITE-Seq data. Samples were then combined into one object and integrated using reciprocal PCA (RPCA) in Seurat using the FindIntegrationAnchors and IntegrateData functions with default parameters. This combined object was then scaled, a PCA was computed, and Louvain clustering was performed with visualization using UMAP with the first 30 principal components (PCs).

CODEX cell type annotation—A flow chart depicting our annotation strategy is provided in Figure S5A. We employed Louvain clustering in Seurat with multilevel refinement (algorithm=2) to determine the cell types present in the CODEX data. Clusters in the integrated Seurat object containing all 12 normal bone marrow (NBM) samples were annotated based on the marker expression of each cluster. Clusters that could not be easily annotated were subjected to a second round of subclustering of just those ambiguous clusters and were resolved. To address the issue of lateral spillover (i.e., bright expression from one cell type may “spill” into its neighbor thereby causing that cell to be mislabeled), we subjected each cluster to an additional round of subclustering, after which we visualized the subclusters on the image itself using cell phenotype maps in Napari. Clusters which represented lateral spillover artifacts were corrected to the appropriate cell type. Lastly, to ensure maximum annotation accuracy, any cell labels which had been changed through this correction process were subjected to one final round of subclustering, where spillover artifacts, if present, were corrected by relabeling the mislabeled subcluster appropriately. In cases where we did not observe a clear cluster for a certain cell type we knew should

be present, we used manual gating to distinguish them, such as immunophenotypic HSCs, CLPs, and GMPs. We identified cells which stained positive for most markers which we used mast cell tryptase (MCT) to visualize because the staining frequency is inconsistent with the relative scarcity of bone marrow mast cells of $\sim 0.01\%$ ⁸⁶, and we thus believed these cells to have highly artifactual staining. We then removed cells positive for MCT and many incompatible markers simultaneously from analysis. Based on the H&E staining, it is possible these were eosinophils, but without a confirmatory marker we considered it safest to simply remove them as they are still relatively rare. We also removed obvious artifacts which were not cells. Other likely artifacts were noted, including CD117 staining in lymphoid cells, CD19 in T-Cells, MCT staining in erythroid cells, and CD10 staining in erythroid cells, CD138 staining in a number of cell types. Some antibodies bound RBCs, especially in AML/NSM samples, but this did not confound analysis as RBCs were annotated easily by GYPC/CD71. ADIPOQ was not specific and thus excluded from downstream analysis. In total, 90.6% of segmented objects were annotated as a specific bone marrow cell type. We detail the markers and strategy used for cell type annotation in Table S4 and the Supplemental Material file called “Cell Type Annotation Guide.” The source code documents the clustering/subclustering process in a step-by-step manner. Cells were assigned a coarse (level 1) and detailed (level 2) type annotation. This process is further detailed in the source code.

We considered how this CODEX cell phenotyping might compare to our scRNA-Seq atlas. While differences in protein and RNA expression and their dynamic ranges make subtle cell type correlation challenging, we did find that when we harmonized the cell labels based on the two modalities, the CODEX-based cell annotations mapped reasonably to the scRNA-Seq-based cell annotations. However, as it is difficult to directly link CODEX data to scRNA-Seq data on a very fine-grained level (for instance, the difference between a GMP and an early myeloid progenitor), we proceeded by keeping the CODEX and scRNA-Seq annotations separate for discussion purposes, while retaining the harmonized labels in the Seurat object metadata.

Analysis of distance between Osteo- and Fibro-MSCs and bone—Due to the bone detachment issues, we were not able to systematically identify these two MSC populations. However, manual inspection of the images revealed cells with the expected immunophenotype of Osteo-MSCs in rare instances of the bone staying on the slide, or in the case of Fibro-MSCs, we found very rare PDPN+ CXCL12+ MCT- cells exclusively in the middle of the region where the trabecular bone used to be. Therefore, Osteo- and Fibro-MSCs were manually annotated in the images (64 Fibro-MSC and 422 Osteo-MSC across all 12 samples). Segmentation masks for each manually annotated binary mask were constructed using a custom Python script, and protein expression was quantified for each cell as described above yielding a cell x protein matrix for each mask. These masks were concatenated and joined with the raw data in full CODEX Seurat object. The same markers were carried forward with the addition of CD56, which was added back to the analysis for all cells. This merged data set was renormalized using centered log ratio (CLR) normalization for each feature independently (margin=1) as above.

Adipo-, THY1+, Osteo-, and Fibro-MSc centroids were extracted from the Seurat object. Then distance to bone was calculated as described above with some modifications. The bone contour points were used as described above, but without downsampling. Given that Osteo- and Fibro-MSCs were frequently observed to occur within regions that contained trabecular bone, Osteo- and Fibro-MSc centroids that were found to overlap the bone annotations were added to the bone contour point list in order to correctly calculate a distance of zero for those cells. Non-normalized distance of each cell to bone was subsequently plotted, and significance between groups was calculated using a Wilcoxon rank-sum test in R.

Correlating cell type annotations by scRNA-Seq and CODEX data—To generate the scRNA-Seq and CODEX correlative plot (Figure S5F-G) to check the relationship between the two sets of cell type annotations, we first generated harmonized cell type labels, representing a set of simplified and common cell type labels we were confident in. We then calculated the average normalized expression of harmonized cell types in both the scRNA-Seq (log-normalized) and CODEX (CLR) data. We also averaged mean expression across cell types in cases where multiple higher-level cell types mapped to a single coarser cell type. Gene names were mapped to corresponding protein names when possible, yielding 49 matching protein/gene pairs to do the analysis. We then computed the Pearson correlation of the min-max normalized (within cell type) expression profiles of a scRNA-Seq and a CODEX-based cell type. The highest matching scRNA-Seq cell types to each CODEX cell type were extracted. We kept the scRNA-Seq profile for osteoblasts labeled as such to help compare to the CODEX “endosteal” label.

CODEX neighborhood analysis—Neighborhood analysis was performed as previously described⁸⁷ using the level 2 cell type annotations for NBM, AML, and NSM samples. Note that AML samples had the additional cell type label of *NPM1* mutant blasts which was not represented in either NBM or NSM samples. The 10 nearest neighbors of each cell were grouped into windows, which were clustered using K-means clustering with K=15 to get neighborhood assignments. Statistical significance of cell type enrichment within each neighborhood was calculated using a hypergeometric test. The hypergeometric test p-value is calculated based on the following four numbers: (1) the number of cells of a given type in the neighborhood; (2) the total number of cells in the neighborhood; (3) the number of cells of a given type in the CODEX dataset; and (4) the total number of cells in the CODEX dataset. P-values were adjusted for multiple hypothesis testing using the Benjamini-Hochberg method and significance was defined as p-adjusted < 0.05⁸⁷.

Integration of CellChat and CODEX neighborhood analyses—To integrate the CellChat and CODEX neighborhood analysis results, we needed a measure which reasonably captures the association of two cell types in a statistically consistent fashion. Given our neighborhood analysis already yielded these associations, we first calculated a neighborhood co-localization score for a given cell type pair i and j in neighborhood k , $NCLS_{ij}^k$, as following:

$$NCLS_{ij}^k = \frac{e_{ij}^k - \overline{e^k}}{\sigma e^k}$$

(1)

where e_{ij}^k is the joint fold enrichment of cell type i and cell type j in neighborhood k which is defined as the sum of the two individual fold enrichment values, calculated as previously described⁸⁷. And $\bar{e}^k$ and σ^k are the mean and standard deviation of the joint fold enrichment of all cell type pairs in the given neighborhood. $NCLS_{ij}^k$ represents how enriched two cell types are in a given neighborhood relative to all other cell type pairs in that neighborhood. The maximum of all $NCLS_{ij}^k$ values ($\max NCLS_{ij}^k$) indicates the maximum co-enrichment of the two cell types across all neighborhoods. Next, we devised a combined measure for prioritization of L-R predictions by simultaneously considering CODEX-derived cell type co-localization score with the CellChat L-R predictions, called CellChat-CODEX Effect Size, $CCES_{ij}$ for cell type i and cell type j . $CCES_{ij}$ is defined as following:

$$CCES_{ij} = \sqrt{\left(\frac{s_{ij} - \min(s_{ij})}{\max(s_{ij}) - \min(s_{ij})}\right) * \left(\frac{\max NCLS_{ij}^k - \min(\max NCLS_{ij}^k)}{\max(\max NCLS_{ij}^k) - \min(\max NCLS_{ij}^k)}\right)} \quad (2)$$

where S_{ij} is the aggregate L-R pair prediction weight by CellChat³⁶. We reasoned that this approach gives users a helpful single metric by which to prioritize L-R predictions which are also associated with co-localization in neighborhoods. We tabulated $NCLS_{ij}^k$ and $\max NCLS_{ij}^k$ scores along with the CellChat L-R predictions in Supplemental Table S3. In cases where a CODEX cell type mapped to multiple scRNA-Seq cell types, we averaged the $NCLS_{ij}^k$ values.

In addition, we calculated the significance of pairwise cell-cell interactions in the CODEX data using a permutation test. Specifically, we used the `testInteractions` function of the `imcRtools` R package (v1.4.2)⁸⁸ with `method="classic"` in which the total number of interactions between cell type i and cell type j (defined by a distance of ≤ 100 pixels ($\sim 51 \mu\text{m}$) between cell centroids) is divided by the total number of cells of type j . Cell labels were permuted 500 times to generate a null distribution of the interaction. Probabilities that the cells were co-localized greater than expected by chance were aggregated using Stouffer's method and recorded in Supplemental Table S3 for each cell type combination.

CODEX reference mapping and validation of predicted ligand-receptor

interactions—CODEX data generated using the new antibody panel including antibodies against the predicted ligand and receptor proteins were computationally processed the same way as NBM samples until the cell type annotation step. Instead of using clustering and manual annotation, we used automated label transfer using RPCA in Seurat v5 with default parameters to map the L-R CODEX datasets onto our NBM atlas and predict their cell labels using the 21 overlapping features ($n = 6$ matching samples; H10, H14, H26, H32, H38, H41). Briefly, each dataset was individually normalized with CLR normalization for each feature independently (`margin=1`), scaled, and a PCA was computed. These objects were then computationally merged, and cells were automatically annotated using

the FindTransferAnchors function with the top 30 dimensions and k.filter=NA, followed by the MapQuery function with default parameters.

Accuracy of reference matching for cell types of interests was evaluated by manually checking the annotation masks on the images. Distance analysis was conducted using the imcRtools R package⁸⁸ with the minDistToCells function to compute the minimum distance from each cell to the source cell type of interest. Due to their rarity and lack of the TDT antibody in the original CODEX dataset used for reference mapping, CLPs were manually annotated in QuPath using CD34+ and TDT+ double positivity and then matching cell IDs were found in the Seurat object and downstream calculations proceeded as with the other cell types. The receptor expression on the receiving cell type was then stratified by quartile based on the CLR-normalized expression within that cell type (0-25%, 25-75%, 75-100%), and data above the 99th percentile of cell distances in each level were filtered prior to plotting. Distance data from the 6 samples were pooled and the minimum distances to the source cell type were compared using a one-sided Wilcoxon Rank Sum test.

NPM1 mutant cell classification and bone distance calculation—NPM1 mutant cells were identified using an Random Trees (RT) classifier in QuPath⁸⁹. Mutant cells were defined as those which expressed NPM1C but not CD141, as CD141 mature myeloid cells stained with the NPM1C mutant-specific antibody even in negative staging marrow controls. Wild type (WT) cell annotation in diagnostic AML samples was aided with visualization of lymphoid markers such as CD3E or mesenchymal markers such as FOXC1. Approximately 100 manually identified mutant and WT cells each were annotated using the brush tool. The cell segmentation mask defined by Mesmer was imported using custom scripts. Using QuPath, for each marker we calculated mean, standard deviation, min, max, Haralick Features (min-NaN, max-NaN, distance-1, bins-32), and the default QuPath morphologic features for each cell detection object. We then trained and applied an RT classifier to all cells from 5 samples and assigned a binary classification of NPM1_Mutant or NPM1_WT to each cell. Classification was manually evaluated in collaboration with clinical hematopathologists, and additional informative annotations were added in an iterative, human-in-the-loop fashion until the classification achieved optimal performance. Detection measurements were exported and merged with the default Seurat analysis by matching cells to their nearest spatial coordinates. We filtered out any leukemic cells which mapped to incompatible (e.g., lymphoid, mesenchymal) or very rare cell types. Spatial analysis with respect to bone was performed using the spatstat R package using the same approach as the healthy neighborhood distance to bone analysis.

Negative staging marrow and AML cell type classification—The same panel was used for AML/NSM CODEX as the femoral head samples, with the exception of CD19 being swapped for NPM1C. AML and NSM samples were computationally processed the same way as NBM samples until the cell type annotation step. Instead of using clustering and manual annotation, we used automated label transfer using RPCA in Seurat to map the NSM and AML datasets onto our NBM atlas and predict their cell labels as described above. Each sample was individually processed (CLR normalization, scaling, PCA) and then projected onto the NBM reference using the FindTransferAnchors function with the top 30

dimensions and $k.filter=NA$ followed by the MapQuery function with default parameters. For comparison of NSM and AML myeloid classification proportions, we considered cells which mapped to GMP, GMP/Myeloblast, Early Myeloid Progenitor, Intermediate Myeloid, Monocyte, Non-Classical Monocyte, pDC, or Macrophages. NSM samples were chosen as the control to account for differences between sample site (femoral head versus iliac crest), sample processing, and generalized stress associated with having cancer.

Annotation of microanatomical structures—Microanatomical structures for proximity analysis were manually annotated. Bone was manually annotated based on morphology and a combination of CD56 expression, exclusion of vessels, and correlation with the corresponding H&E image. Adipocytes were annotated using CD146 and morphology, with VE-Cadherin-positive arterioles and sinusoids excluded. Structures with characteristic ASMA+ cells surrounding a layer of VECAD+ CXCL12+ endothelial cells were labeled as arterioles, while vessels with VECAD/CD34 expression but not ASMA or CXCL12 were labeled as sinusoids. CXCL12+ stroma and CD163+ macrophages were annotated using manual thresholding on a per-image basis in QuPath. Macrophages were considered a structure due to their considerable cytoplasmic projections which were not well captured by traditional cell/nuclear segmentation-based approaches. Stroma, macrophages, arterioles, and bone were annotated across each entire image. Adipocytes and sinusoids were prohibitively abundant to annotate across the entire image, and thus large ROIs with hundreds of each structure were annotated per each image.

Structural proximity analysis—We measured the proximity to microanatomical structures (adipocyte, arteriolar, sinusoid, bone, macrophage, stroma) from different cell types, neighborhoods and structures using the median distance and applying a Poisson point process model. The key properties of the Poisson point process include the independence of events within non-overlapping regions and the probability of occurrence of events following a Poisson distribution. This makes the Poisson point process particularly useful for modeling rare cells.

$$\lambda(u) = \exp(\beta_0 + \beta_1 S(u)) \quad (3)$$

The model is applied to our data where u is the x-y coordinate of a cell, $\lambda(u)$ is probability of a cell, β_0 is the population of cells to optimize, β_1 is decay rate of $S(u)$ to optimize, $S(u)$ is distance to the structure. The value of $\lambda(u)$ is 0 when there are no cells at position u , and 1 if there is a cell. When fitting the data, $\lambda(u)$ ranges from 0 to 1. β_0 is highly correlated to the population size, which can be counted directly. And β_1 is what we are interested in, it represents the proximity. A small β_1 value indicates that the distance is close. The Poisson point process model is implemented in the spatstat R package. We used the “ppm” function from that package to calculate the β_0 and β_1 . We quantified spatial proximity using annotation mask images and a cells table that includes cell locations and labels (cell types and neighborhoods). To extract contour points for the structures, we used the cv2 package in Python. We then randomly sampled 10% of these points to create a

spatial point pattern for the structures. To prevent overemphasis of outliers, we excluded rare cell types from our analysis. Specifically, if a cell type has a population of less than 10 in a given sample, it is excluded from that sample's rank calculation. Additionally, if there are fewer than 6 samples for a particular cell type after all calculations have been completed, that cell type is also excluded from our analysis. We utilized rank proximity as our final metric, rather than physical distance, because the sizes of different marrow cavities may differ, and an internally normalized metric was more appropriate. To determine the proximity of cell types, neighborhoods or structures to structures, we used a normalized rank calculation. This involved ranking the β_i value for cells in comparison in every sample and scaling the ranking number to a range of 0-1 within each sample. We conducted a one-sided permutation test without replacement to determine if the cell types, neighborhoods, or other structures are significantly close to the target structure. We permuted the cell type labels 100 times and calculated their median distance across the 100 permutations. The proportion of distances smaller than that of the cell type is the p-value. We used Stouffer's method for the calculation of p-values for cell types aggregated from the permutation test p-values.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Highlights

- scRNA-Seq and CODEX reveal the composition and spatial architecture of human bone marrow
- Human mesenchymal stromal cells are transcriptionally and functionally heterogeneous
- Adult HSPCs exhibit a peri-adipocytic spatial localization
- CODEX reference mapping to our healthy atlas reveals MSC expansion in AML

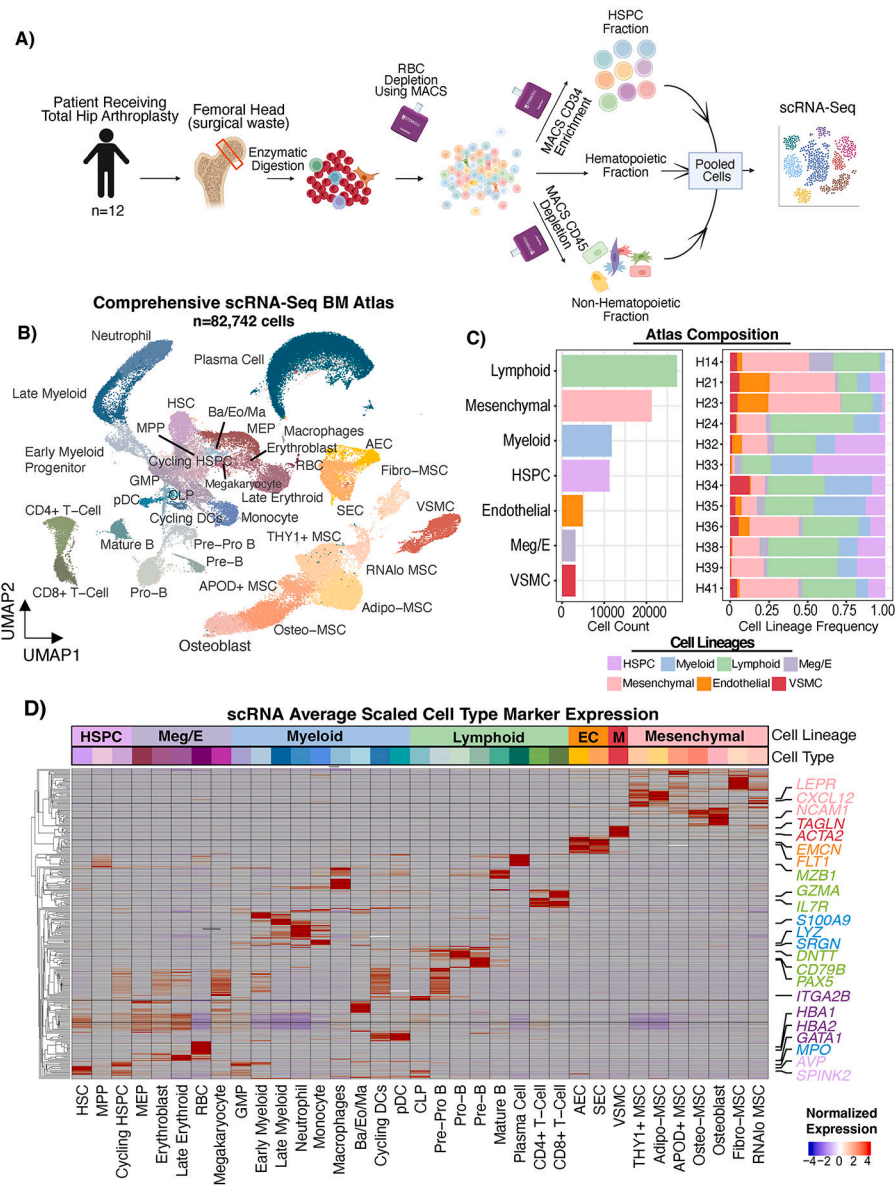


Figure 1 – A Single-Cell transcriptomic atlas of hematopoietic and non-hematopoietic cells of human bone marrow.

A) Schematic for the scRNA-Seq workflow. Magnetic-activated cell sorting (MACS) separation of hematopoietic, stem/progenitor, and mesenchymal fractions was performed and then pooled into one scRNA-Seq reaction per patient. **B)** UMAP representation of 82,742 single-cell transcriptomes from bone marrow of 12 individuals. AEC, arterial endothelial cell; SEC, sinusoidal endothelial cell; VSMC, vascular smooth muscle cell; Ba, Basophil; Eo, Eosinophil; Ma, Mast Cell; RBC, red blood cell; pDC, plasmacytoid dendritic cell; CLP, common lymphoid progenitor; MEP, megakaryocyte erythroid progenitor; GMP, granulocyte monocyte progenitor; MPP, multipotent progenitor; HSPC, hematopoietic stem and progenitor cell; HSC, hematopoietic stem cell; Meg/E, megakaryocyte/erythroid; MSC, mesenchymal stromal cell. **C)** Bar plots showing the cell counts for each lineage captured (left) and the cell lineage proportions per sample (right). **D)** Heatmap showing normalized

gene expression scaled by row (gene) of top differentially expressed and key cell lineage marker genes. EC, endothelial cell; M, vascular smooth muscle. Genes are color-coded to match the lineage of the cell type in which they are differentially expressed.

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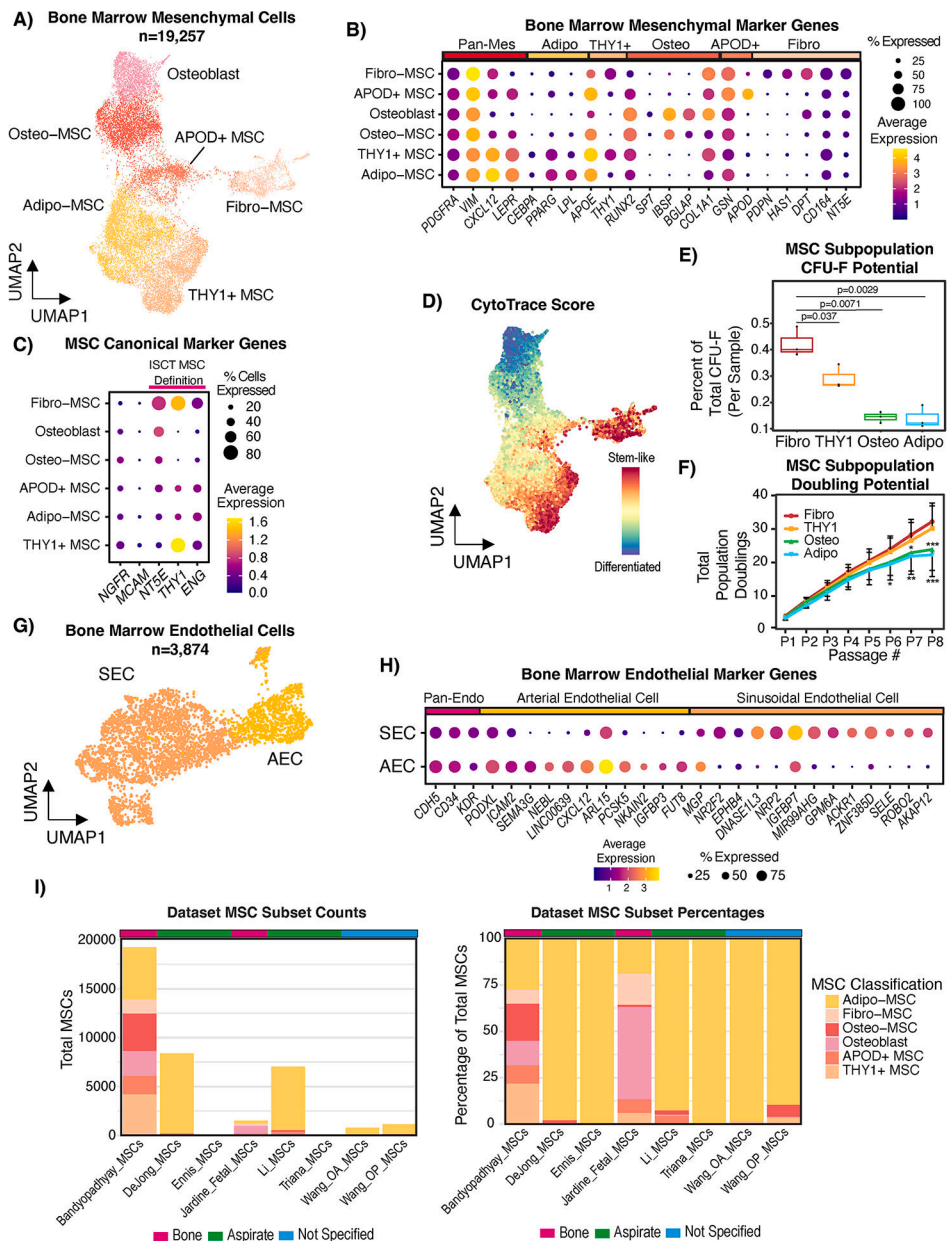


Figure 2 – Defining the non-hematopoietic cellular composition of human bone marrow.

A) UMAP of 19,257 mesenchymal cells from 12 individuals showing different mesenchymal subsets, with RNAlo MSCs being excluded due to their QC profile. **B)** Dot plot showing normalized expression of key mesenchymal marker genes in MSC subsets. **C)** Dot plot showing the normalized expression of literature-derived marker genes for human MSCs including *NT5E* (CD73), *THY1* (CD90), and *ENG* (CD105). *NGFR* (CD271)²⁵ and *MCAM* (CD146)⁹⁰ have also been described as canonical MSC markers. **D)** CytoTRACE analysis projected onto the MSC UMAP showing the predicted differentiation score. **E)** Boxplots showing the relative fibroblast colony forming potential of sorted MSC subtypes. Each data point is one sample. P-values were computed using Welch's two-sample two-sided test. **F)** Line plot showing the population doublings over the course of cell culturing

for eight passages, where all cells were passaged every 7 days. P-values were computed as Fibro-MSCs vs other MSCs by two-way ANOVA. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$. **G)** UMAP showing 3,874 endothelial cells from the 12 individuals. SEC, Sinusoidal Endothelial Cells; AEC, Arterial Endothelial Cells. **H)** Dot plot showing normalized expression of selected pan-endothelial and differentially expressed genes by adjusted p-value between endothelial subsets. **I)** Reference mapping of MSC subsets from human bone marrow between this study and published studies. Left, counts of mapped MSCs in each dataset; Right, percentages of mapped MSCs in each dataset. Datasets are color-coded based on the experimental protocol.

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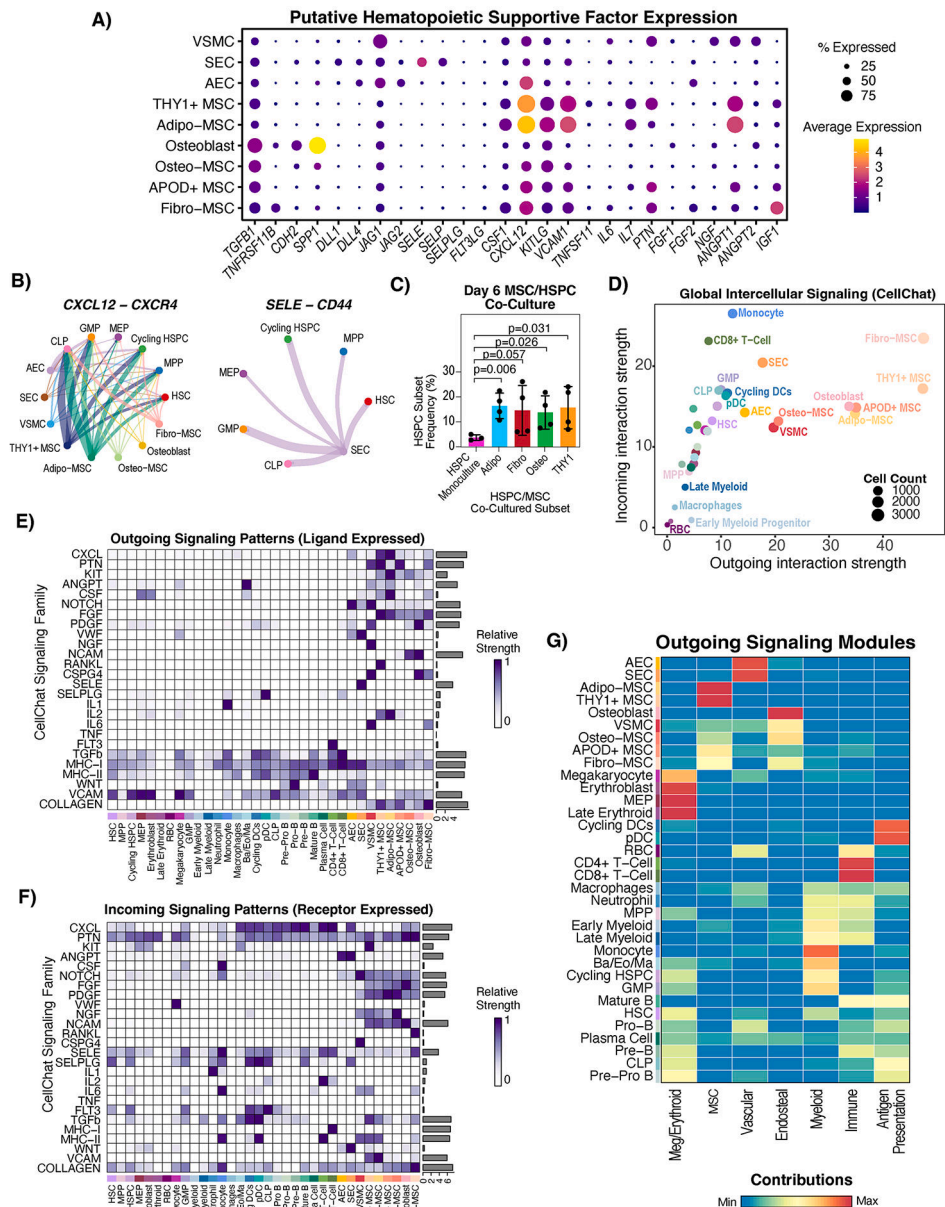


Figure 3 – Cell-cell communication analysis reveals diverse signaling patterns between hematopoietic and non-hematopoietic cell types.

A) Dot plot showing the normalized expression of manually curated support factors for hematopoiesis in mesenchymal and endothelial populations. **B)** Chord diagrams showing significant interactions between source mesenchymal/endothelial cell types and target HSPC cell types predicted by CellChat. Thickness of the line corresponds to the strength of the predicted interaction. **C)** Co-culture of freshly isolated CD34+ HSPCs with sorted and cultured MSC subsets. Error bar represents mean $\pm$ SD from samples from 3 (monoculture) or 4 (co-culture) independent experiments. P-values were computed using Welch's two-sample one-sided test. **D)** CellChat network analysis with cell types scored based on their outgoing and incoming contributions to the network, where strength is defined by number

of significant outgoing interactions and the count refers to the number of cells in each group. **E-F**) Outgoing (ligand enriched, E) and incoming (receptor enriched, F) significant signaling predictions were scored for each annotated signaling family, and the strength of signaling (CellChat interaction probability) for each cell type was row-scaled and plotted by pathway. The barplot on the right-hand side of each panel shows the total signaling strength across all cell groups. **G**) Non-negative matrix factorization (NMF) to identify modules of outgoing signaling and the contribution of each cell type to the pattern.

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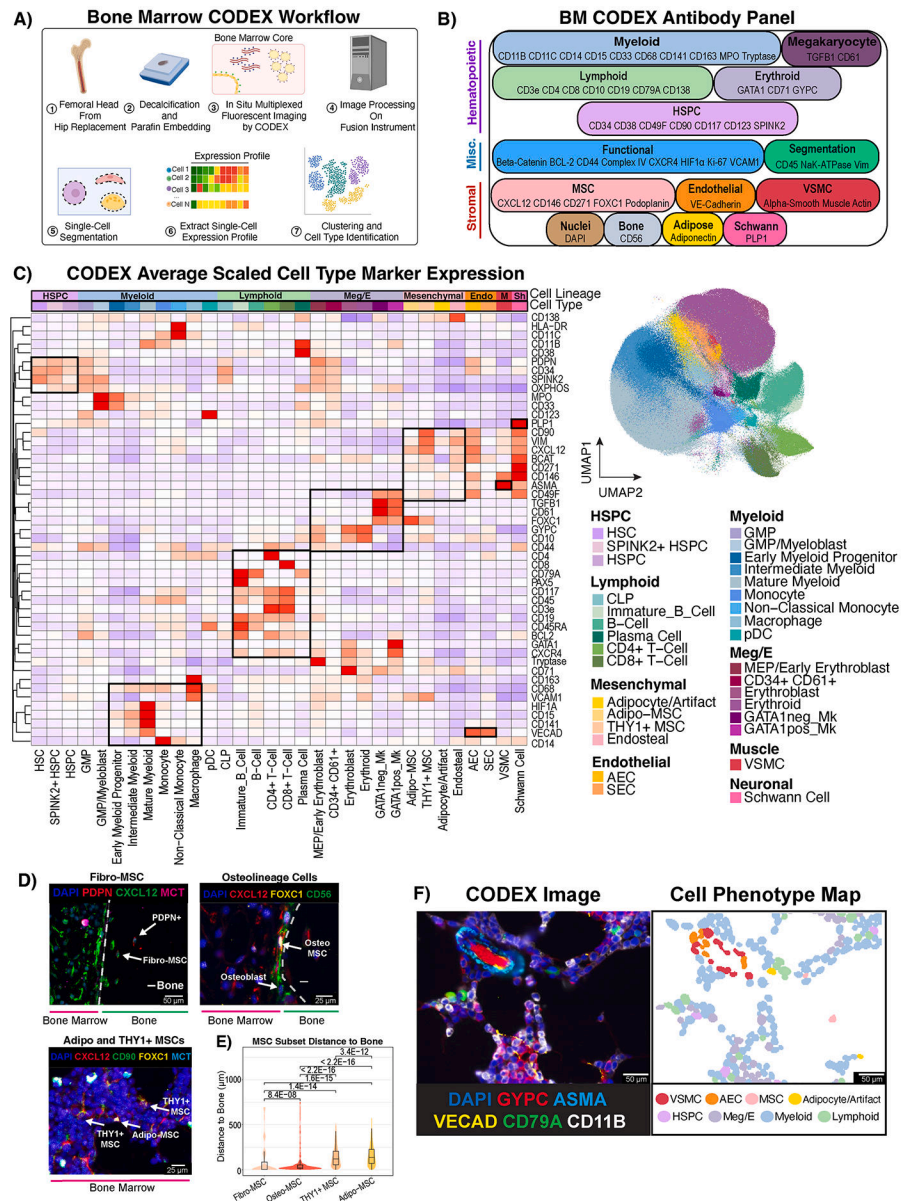


Figure 4 – 54-plex CODEX imaging reveals the spatial cellular topography of human bone marrow.

A) Schematic depicting the CODEX experimental and computational workflow leading to cell type identification. **B)** 53-antibody CODEX panel split by target cell population. **C)** Heatmap showing average CLR-normalized expression per cell type scaled by protein marker and hierarchically clustered by protein expression (left). Boxes highlight coordinate marker expression. UMAP showing the 803,131 single cells in the CODEX atlas from 12 individuals colored by cell type (right). Sh, Schwann Cells. **D)** CODEX images showing manually identified examples of Fibro-MSC, osteolineage cells (osteo-MSC (CXCL12+ CD56+), osteoblast (CXCL12lo CD56+)), Adipo-MSCs (FOXC1+ CXCL12+) and THY1+ MSCs (CD90+ CXCL12+ FOXC1+). **E)** Violin plot showing the nearest distance of annotated cells from different MSC subtypes (n=64 Fibro-MSC, n=422 Osteo-MSC,

n=5,110 Adipo-MSc, and n=4,108 THY1+ MSc) to the manually annotated bone contours. Distances were clipped to the 99th percentile for each cell type for improved visualization. P-values were computed using two-sided Wilcoxon rank sum test. **F**) CODEX image (left) is paired with the cell phenotype map (CPM, right), showing the segmented cells colored by cell annotation.

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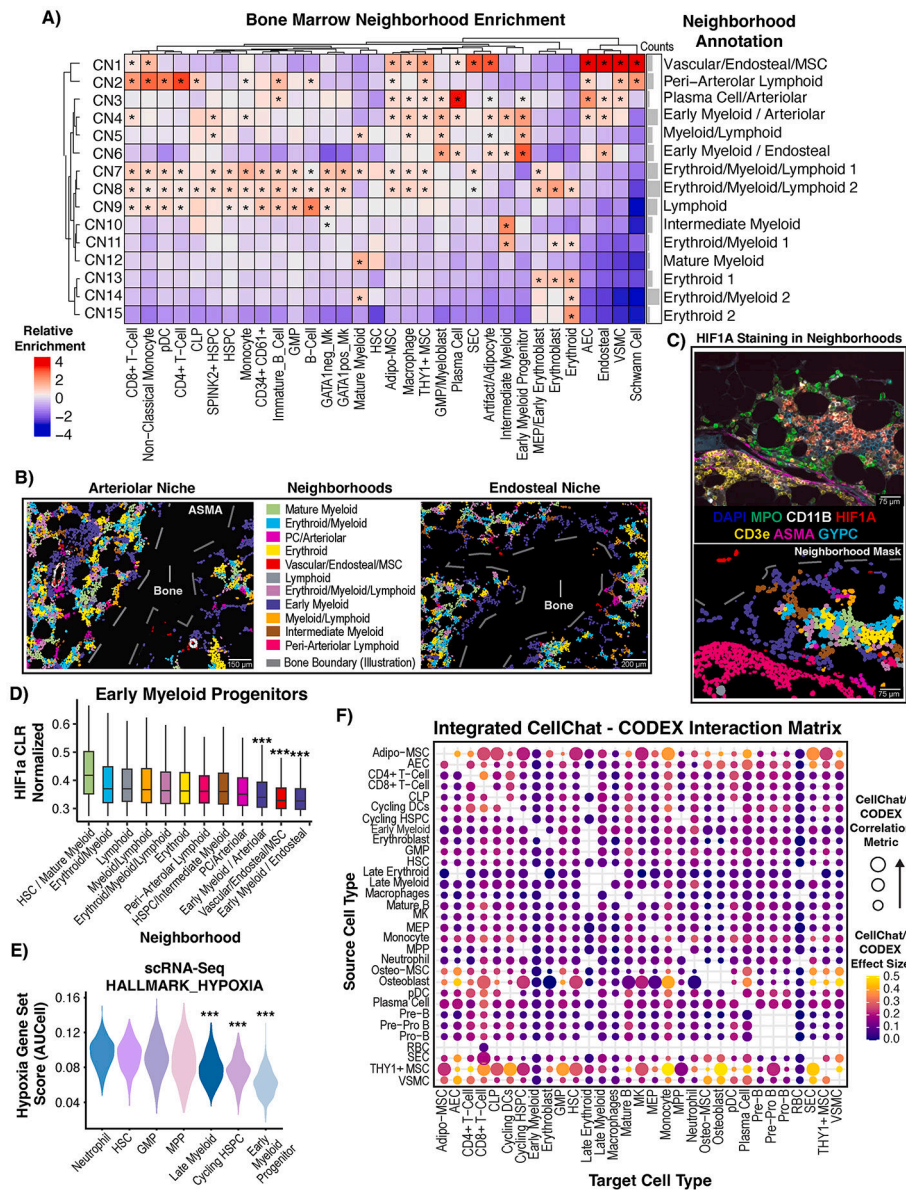


Figure 5 – Neighborhood analysis reveals an oxygen-rich arteriolar/endosteal niche for GMP/early myeloid progenitor cells.

A) Heatmap showing enrichment of the cell types present in neighborhoods. Neighborhood names were assigned based on the enriched cell types. P-values for enrichment were computed using a hypergeometric test and adjusted for multiple hypothesis testing using the Benjamini-Hochberg method. *, $p < 0.05$. **B)** Cell phenotype masks colored by neighborhood membership were plotted alongside ASMA fluorescent signal (left only). **C)** HIF1A staining pattern across neighborhoods. Top, CODEX image showing the MPO⁺ early myeloid progenitors not expressing HIF1A, in contrast to mature myeloid elements. Bottom, neighborhood masks which share a legend with Panel B. **D)** Boxplots showing the normalized HIF1A expression levels in early myeloid progenitors split by neighborhood membership. Data were clipped between 0.1 and 0.95 for improved visualization only. P-values were computed using Welch's one-sided two-sample test comparing cells from

each neighborhood to cells in all other neighborhoods and adjusted using the Benjamini-Hochberg method. *, $p < 0.05$; ***, $p < 0.001$. **E)** Violin plots showing hypoxia signature scores computed using AUCell and our scRNA-Seq data. P-values were comparing AUCell hypoxia scores from each cell type to all other cell types as in (D). **F)** Bubble plot showing correlation and CODEX-CellChat effect size (CCES) between CellChat interaction prediction strength and max CODEX spatial neighborhood co-localization strength of two cell types.

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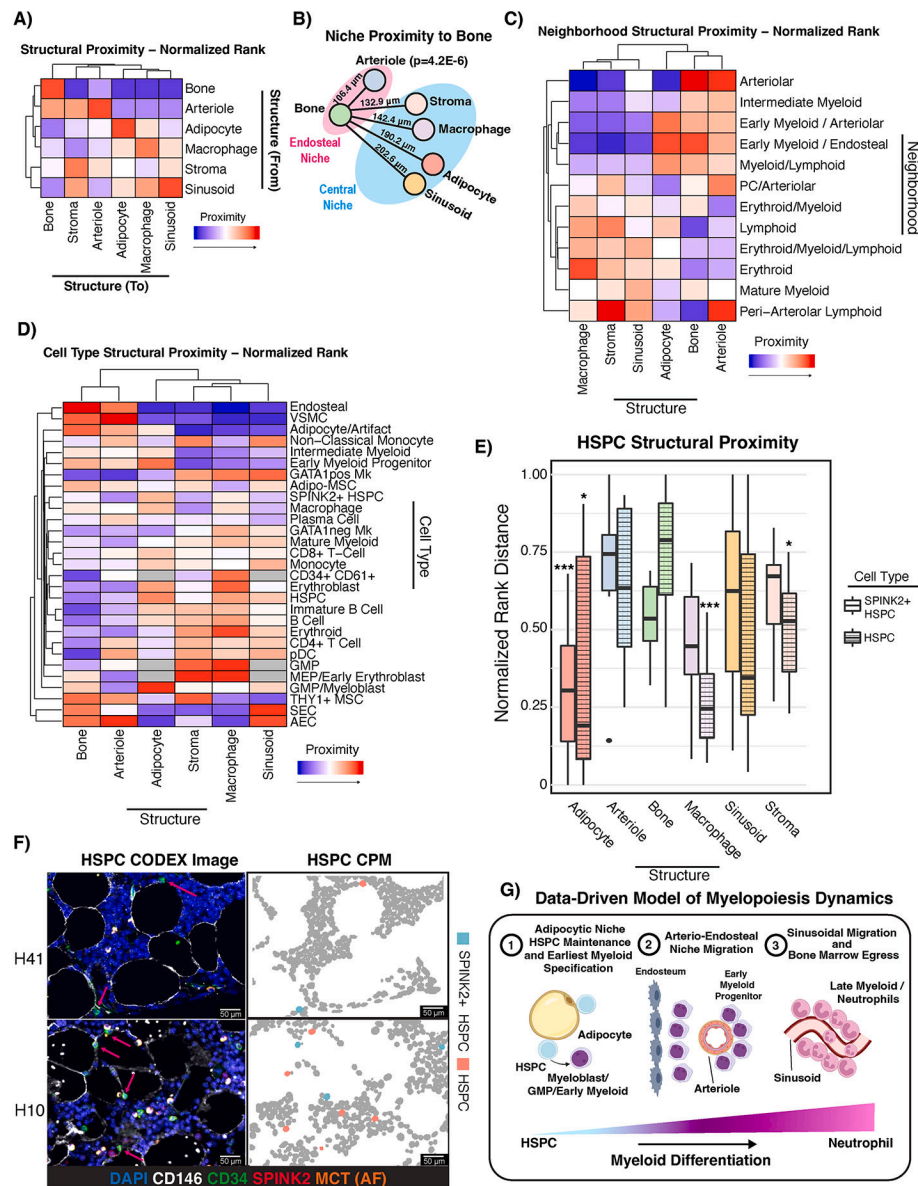


Figure 6 – Comprehensive structural analysis of bone marrow microenvironment uncovers adipocytic localization of lineage-unspecified HSPCs.

A) Heatmap showing the normalized rank proximity from each microenvironmental structure to each structure, using a Poisson point process model ($n=12$). Each structure was ranked by its proximity to each other structure in each sample, and the rank was normalized such that a value of 0 (red) means most proximal and 1 (blue) means least proximal.

B) Data-driven illustration of structural proximity in normal bone marrow. Edge lengths were scaled to the median normalized rank proximity of each structure, and the physical median distance across 12 samples was labeled above each line. The only significant p -value is highlighted, which was computed using Stouffer's method to aggregate per-sample permutation test p -values. **C-D)** Heatmap showing the normalized rank proximity of each neighborhood (C) and cell type to each type of microenvironmental structure (D). Data is from all 12 samples, except when a cell type or neighborhood was too rare to rank in a

sample (<10 in at least half of samples, indicated by grey color or excluded from heatmap). **E)** Boxplot showing the normalized rank proximity of SPINK2+ HSPCs (solid boxes) and HSPCs (dashed boxes) to each microenvironmental structure. P-values were computed by Stouffer's method as in (B) comparing observed vs. random distance of each HSPC subset to each structure. *, $p < 0.05$; ***, $p < 0.001$. **F)** Representative CODEX images and associated cell phenotype maps from two samples showing the peri-adipocytic localization of both HSPCs and SPINK2+ HSPCs. **G)** Cartoon showing a proposed model of human myelopoiesis spatial dynamics.

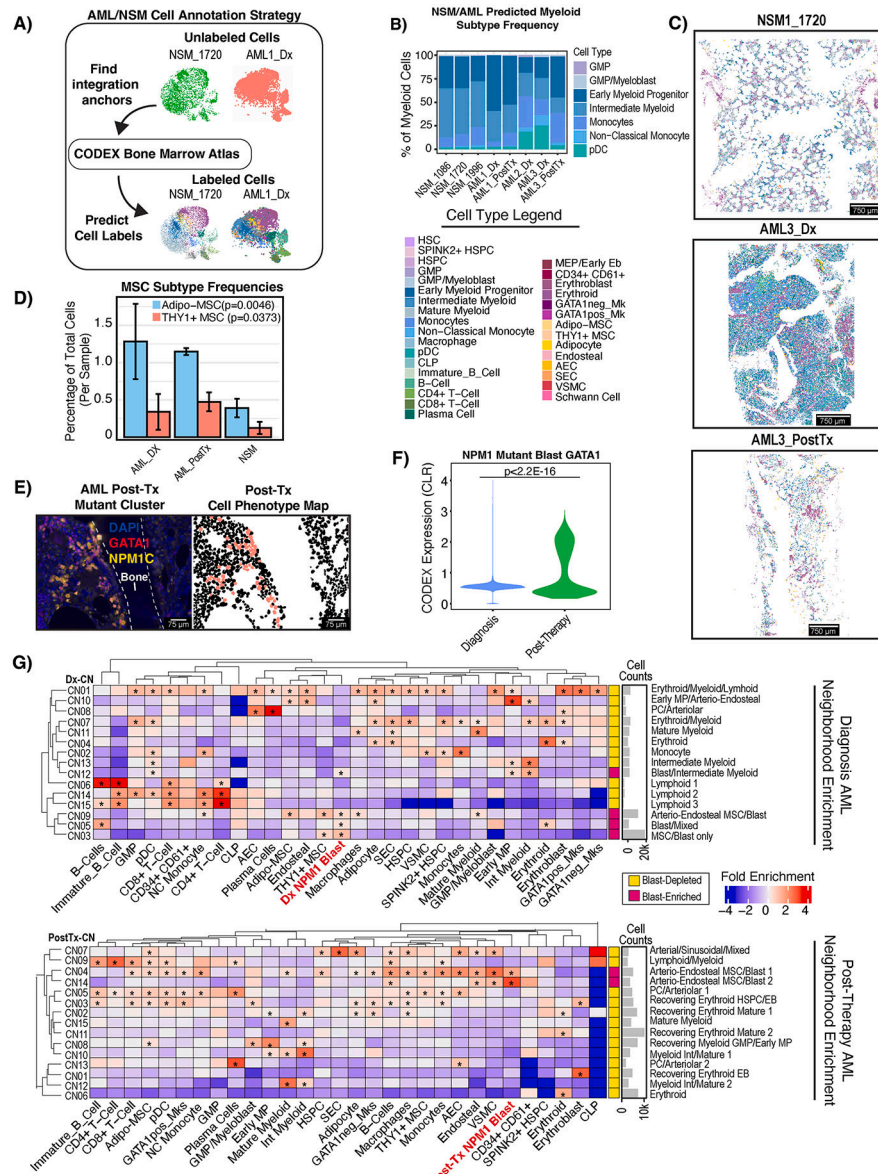


Figure 7 – Unsupervised single-cell mapping of AML reveals stromal expansion and MSC-enriched AML-specific neighborhoods.

A) Schematic showing our unsupervised label transfer computational approach. **B)** Bar plot showing the myeloid cell type frequencies in each sample, excluding mature myeloid cells. **C)** Cell phenotype maps (CPMs) shown for representative sample images. Masks are colored by cell type and share a legend with (B). Dx – Diagnostic, PostTx – Post-Therapy. **D)** Bar plot showing per sample frequency of MSCs in AML versus NSM. MSC frequency was calculated as the proportion of cells in each sample which were annotated as Adipo- or THY1+ MSCs. The distribution of MSC subtype frequencies in AML samples (AML_Dx and PostTx combined) was compared to that of NSM samples using Welch's two-sample two-sided test. **E)** CODEX image showing clustering of rare residual *NPM1* mutant blasts which also stained positive for GATA1, juxtaposed with the CPM highlighting the segmented blasts. **F)** Violin plot showing GATA1 protein levels in the *NPM1* mutant blasts

from diagnosis and post therapy samples. P-value was computed using Welch's two-sample two-sided test. **G**) Heatmap showing the neighborhood cell enrichments (AML: n=175,365 cells, NSM: n=152,198 cells). P-value for enrichment was computed using hypergeometric test and adjusted for multiple hypothesis testing using the Benjamini-Hochberg method. *, $p < 0.05$.

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Key resources table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit polyclonal anti-alpha smooth muscle Actin	Abcam	Cat# ab5694, RRID:AB_2223021
Mouse monoclonal anti-CD19 (clone MRQ-36) custom carrier-free	Cell Marque	Cat# CD19, RRID:AB_3083084
Anti-Hu CD4 (AKYP0048)-BX003-Alexa Fluor™ 647 for PhenoCycler	Akoya Biosciences	Cat# 4350018, RRID:AB_2915936
Mouse monoclonal anti-CD71 (clone MRQ-48) custom carrier-free	Cell Marque	Cat# CD71, RRID:AB_3083088
Anti-Hu CD44 (AKYP0073)-BX005-Atto 550 for PhenoCycler	Akoya Biosciences	Cat# 4450041, RRID:AB_2936081
Rabbit recombinant monoclonal anti-CD271 (clone EP1039Y)	Abcam	Cat# ab256584, RRID:AB_3076736
Mouse monoclonal anti-Vimentin (clone RV202)	Novus Biologicals	Cat# NBP1-97672, RRID:AB_2915945
Goat polyclonal anti-CD117	R & D Systems	Cat# AF1356, RRID:AB_354750
Rabbit recombinant monoclonal anti-CD61	Abcam	Cat# ab240214, RRID:AB_3076749
Rabbit polyclonal anti-CD90	Cell Signaling Technology	Cat# 16836, RRID:AB_3076750
Anti-Hu CD68(AKYP0050)-BX015-Alexa Fluor™ 647 for PhenoCycler	Akoya Biosciences	Cat# 4550113, RRID:AB_2935894
Mouse monoclonal anti-CD45RA (clone H100)	Biolegend	Cat# 304102, RRID:AB_314406
Rabbit polyclonal anti-GYPC	Thermo Fisher Scientific	Cat# PA5-80680, RRID:AB_2787967
Rabbit recombinant monoclonal anti-CD49F	Abcam	Cat# ab240252, RRID:AB_3077780
Rabbit recombinant monoclonal anti-CXCR4 (clone D4Z7W)	Cell Signaling Technology	Cat# 90927, RRID:AB_3078806
Anti-Hu CD45 (AKYP0074)-BX021-Alexa Fluor™ 647 for PhenoCycler	Akoya Biosciences	Cat# 4450042, RRID:AB_2915946
Mouse monoclonal anti-CD15 (clone SSEA-1)	Biolegend	Cat# 301902, RRID:AB_314194
Rabbit polyclonal anti-SPINK2	Novus Biologicals	Cat# NBP2-98568, RRID:AB_3079877
Rabbit recombinant monoclonal anti-CD11C (clone EP1347Y)	Abcam	Cat# ab216655, RRID:AB_2864379
Rabbit recombinant monoclonal anti-FOXC1 (clone EPR20678)	Abcam	Cat# ab230157, RRID:AB_3080862
Anti-Hu CD8(AKYP0028)-BX026-Atto 550 for PhenoCycler	Akoya Biosciences	Cat# 4250012, RRID:AB_2915960
Mouse monoclonal anti-Adiponectin (clone OT11H1)	Novus Biologicals	Cat# NBP2-71422, RRID:AB_2915961
Mouse monoclonal anti-VE-cadherin (clone 16B1)	Thermo Fisher Scientific	Cat# 14-1449-82, RRID:AB_467495
Rabbit recombinant monoclonal anti-CD123 (clone IL3RA/2947R)	Novus Biologicals	Cat# NBP2-79851, RRID:AB_3080863
Rabbit recombinant monoclonal anti-CD14 (clone SP192)	Abcam	Cat# ab230903, RRID:AB_2889378
Mouse monoclonal anti-OxPhos complex IV subunit IV (clone 10G8D12C12)	Thermo Fisher Scientific	Cat# A-21347, RRID:AB_221508
Rabbit recombinant monoclonal anti-PLP1 (clone EPR23504-106)	Abcam	Cat# ab275751, RRID:AB_2915963
Anti-Hu HLA-DR(AKYP0063)-BX033-Alexa Fluor™ 647 for PhenoCycler	Akoya Biosciences	Cat# 4550118, RRID:AB_3080864
Rabbit recombinant monoclonal anti- Mast Cell Tryptase (clone EPR8476)	Abcam	Cat# ab227572, RRID:AB_3080865
Mouse monoclonal anti-CD34 (clone QBEnd/10)	Abcam	Cat# ab213054, RRID:AB_2811050
Mouse monoclonal anti-CD10 (clone MME/1892)	Abcam	Cat# ab237866, RRID:AB_3080866

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Rabbit recombinant monoclonal anti-Pax5 (clone D7H5X)	Cell Signaling Technology	Cat# 93009, RRID:AB_3080867
Rabbit recombinant monoclonal anti-CD11b (clone EPR1344)	Abcam	Cat# ab209970, RRID:AB_2915959
Rabbit recombinant monoclonal anti-VCAM1 (clone EPR5047)	Abcam	Cat# ab271899, RRID:AB_2895043
Rabbit recombinant monoclonal anti-TGF beta 1 (clone EPR21143)	Abcam	Cat# ab229856, RRID:AB_3080868
Rabbit recombinant monoclonal anti-CD146 (clone EPR3208)	Abcam	Cat# ab210072, RRID:AB_3081922
Anti-Hu CD3e(AKYP0062)-BX045-Alexa Fluor™ 647 for PhenoCycler	Akoya Biosciences	Cat# 4550119, RRID:AB_2936080
Rabbit recombinant monoclonal anti-GATA1 (clone D52H6)	Cell Signaling Technology	Cat# 40977, RRID:AB_3082417
Anti-Hu/Mu Ki67(AKYP0052)-BX047-Atto 550 for PhenoCycler	Akoya Biosciences	Cat# 4250019, RRID:AB_2895046)
Rabbit recombinant monoclonal anti-CD138 (clone EPR6454)	Abcam	Cat# ab216458, RRID:AB_2889392
Mouse monoclonal anti-CXCL12 (clone 79018)	R & D Systems	Cat# MAB350, RRID:AB_2088149
Mouse monoclonal anti-CD33 (clone 996810)	R & D Systems	Cat# MAB11371, RRID:AB_2889385
Rabbit monoclonal anti-CD56 (clone MRQ-42) custom carrier-free	Cell Marque	Cat# CD56, RRID:AB_3082973
Rabbit recombinant monoclonal Anti-Sodium Potassium ATPase (clone EP1845Y)	Abcam	Cat# ab167390, RRID:AB_2890241
Anti-Hu HIF1A(AKYP0100)-BX062-Alexa Fluor™ 647 for PhenoCycler	Akoya Biosciences	Cat# 4550069, RRID:AB_3082972
Anti-Hu CD163(AKYP0114)-BX069-Atto 550 for PhenoCycler	Akoya Biosciences	Cat# 4250079, RRID:AB_2935895
Anti-Hu Bcl-2(AKYP0120)-BX085-Atto 550 for PhenoCycler	Akoya Biosciences	Cat# 4250098, RRID:AB_3082974
Anti-Hu CD141(AKYP0124)-BX087-Atto 550 for PhenoCycler	Akoya Biosciences	Cat# 4250097, RRID:AB_3082975
Anti-Hu CD38(AKYP0110)-BX089-Atto 550 for PhenoCycler	Akoya Biosciences	Cat# 4250080, RRID:AB_3082976
Anti-Hu CD79a(AKYP0109)-BX090-Alexa Fluor™ 750 for PhenoCycler	Akoya Biosciences	Cat# 4450078, RRID:AB_3082977
Anti-Hu b-Catenin1(AKYP0068)-BX096-Atto 550 for PhenoCycler	Akoya Biosciences	Cat# 4250091, RRID:AB_3082978
Anti-Hu MPO(AKYP0113)-BX098-Atto 550 for PhenoCycler	Akoya Biosciences	Cat# 4250083, RRID:AB_2927678
Anti-Hu Podoplanin(AKYP0007)-BX121-Atto 550 for PhenoCycler	Akoya Biosciences	Cat# 4250094, RRID:AB_3082979
Rabbit polyclonal anti-NPM1 (mutant)	Thermo Fisher Scientific	Cat# PA1-46356, RRID:AB_2267471
Rabbit monoclonal anti-TDT (clone EP266) custom carrier-free	Cell Marque	Cat# TDT, RRID:AB_3082981
Rabbit recombinant monoclonal Anti-COL1a (clone E8F4L)	Cell Signaling Technology	Cat# 81375, RRID:AB_3083034
Rabbit monoclonal anti-JAG1 (clone D4Y1R) custom carrier-free	Cell Signaling Technology	Cat # Jagged1, RRID:AB_3083044
Rabbit polyclonal anti-NOTCH3	Abcam	Cat# ab23426, RRID:AB_776841
Rat monoclonal anti-DLL4	R & D Systems	Cat# MAB1389, RRID:AB_2092985
Rabbit monoclonal anti-FGF2 (clone E5Y6M) custom carrier-free	Cell Signaling Technology	Cat# Basic FGF, RRID:AB_3083050
Mouse monoclonal anti-IL7 (clone IL7/4012)	Thermo Fisher Scientific	Cat# 3574-MSM2-P0, RRID:AB_3083054
Rabbit recombinant monoclonal anti-IL7R (clone EPR23747-333)	Abcam	Cat# ab282011, RRID:AB_2915969
Goat polyclonal anti-ANGPT1	Thermo Fisher Scientific	Cat# PA5-46968, RRID:AB_2607762
Rabbit polyclonal anti-TIE2	Novus Biologicals	Cat# NBP1-69753, RRID:AB_1101777

REAGENT or RESOURCE	SOURCE	IDENTIFIER
BD OptiBuild™ BUV395 Mouse Anti-Human CD61 (Clone RUU-PL7F12)	BD Biosciences	Cat# 745717, RRID:AB_2871633
Brilliant Violet 510™ anti-human CD19 (Clone HIB19)	Biolegend	Cat# 302241, RRID:AB_2561381
Brilliant Violet 605™ anti-human CD14 (Clone 63D3)	Biolegend	Cat# 367125, RRID:AB_2716230
Brilliant Violet 650™ anti-human CD33 (Clone WM53)	Biolegend	Cat# 303429, RRID:AB_2650933
Brilliant Violet 785™ anti-human CD38 (Clone HIT2)	Biolegend	Cat# 303530, RRID:AB_2565893
FITC anti-human CD15 (Clone MMA)	Biolegend	Cat# 394705, RRID:AB_2750481
PerCP/Cyanine5.5 anti-human CD235ab (Clone HIR2)	Biolegend	Cat# 306613, RRID:AB_10696551
PE anti-human CD66a/c/e (Clone ASL-32)	Biolegend	Cat# 342303, RRID:AB_1626288
APC/Cyanine7 anti-human CD45 (Clone 2D1)	Biolegend	Cat# 368516, RRID:AB_2566376
Alexa Fluor® 647 anti-human CD163 (Clone GHI/61)	Biolegend	Cat# 333619, RRID:AB_2563474
PE/Cyanine7 anti-human CD34 (Clone 561)	Biolegend	Cat# 343615, RRID:AB_2629725
Alexa Fluor® 700 anti-human CD71 (Clone CY1G4)	Biolegend	Cat# 334130, RRID:AB_2888770
FITC anti-human CD45 (Clone HI30)	Biolegend	Cat# 304038, RRID:AB_2562050
PE/Dazzle anti-human Podoplanin (Clone NC-08)	Biolegend	Cat# 337028, RRID:AB_2750288
PE/Cy7 anti-human CD56 (Clone 5.1H11)	Biolegend	Cat# 362510, RRID:AB_2563927
Brilliant Violet 650™ anti-human CD90 (Clone 5E10)	Biolegend	Cat# 328143, RRID:AB_2734319
APC/Fire anti-human CD235ab (Clone HIR2)	Biolegend	Cat# 306622, RRID:AB_2716166
Alexa Fluor 647 anti-human VE-Cadherin (Clone BV9)	Biolegend	Cat# 348513, RRID:AB_2563669
PE anti-human Leptin Receptor (LepR) (Clone 52263)	R&D Systems	Cat# FAB867P, RRID:AB_2249952
Bacterial and virus strains		
Biological samples		
Human Femoral Head	This Manuscript	NA
Human Acute Myeloid Leukemia FFPE Blocks	This Manuscript	NA
Human Negative Staging Marrow FFPE Blocks	This Manuscript	NA
Chemicals, peptides, and recombinant proteins		
MesenCult Proliferation Kit	STEMCELL Technologies	Cat# 05411
Dispase II	Sigma-Aldrich	Cat# D4693, CAS# 42613-33-2
Collagenase, Type 1	Worthington Biochem	Cat# LS004196
Indomethacin	Sigma-Aldrich	Cat# I7378, CAS# 53-86-1
Isobutylmethylxanthin	Sigma-Aldrich	Cat# I5879, CAS# 28822-58-4
Hydrocortisone	Sigma-Aldrich	Cat# H4001, CAS# 50-23-7
Insulin	Gibco	Cat# 12585014
Dexamethasone	Sigma-Aldrich	Cat# D8893, CAS# 50-02-2
Potassium phosphate monobasic	Sigma-Aldrich	Cat# P5655, CAS# 7778-77-0
TGF-beta1	Sigma-Aldrich	Cat# GF346
Sodium Pyruvate	Gibco	Cat# 11360070
L-Proline	Sigma-Aldrich	Cat# P5607, CAS# 147-85-3
ITS+ Premix	Corning	Cat# 354352

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Alizarin Red S	Sigma-Aldrich	Cat# A5533, CAS# 130-22-3
Alcian Blue	Sigma-Aldrich	Cat# A5268, CAS# 33864-99-2
Crystal Violet	Sigma-Aldrich	Cat# C6158, CAS# 548-62-9
Human TruStain FcX™ (Fc Receptor Blocking Solution)	Biolegend	Cat# 422302
UltraComp eBeads Compensation Beads	Invitrogen	Cat# 01-2222-42
Histochoice clearing agent	VWR	Cat# H103-4L
Assay Reagent for PhenoCycler	Akoya Biosciences	Cat# 7000002
10X Buffer Kit for PhenoCycler-Fusion	Akoya Biosciences	Cat# 7000019
Nuclear Stain for PhenoCycler	Akoya Biosciences	Cat# 7000003
Citrate Buffer, pH 6.0, 10×, Antigen Retriever	Sigma-Aldrich	C9999-1000ML
EasySep™ Buffer	STEMCELL Technologies	Cat# 20144
Paraformaldehyde 16% Aqueous Solution EM Grade	Electron Microcopy Sciences	Cat# 15710-S
UltraPure 0.5M EDTA, pH8.0	Invitrogen	Cat# 15575-038
Formical-4	StatLab	Cat# 1214-1
Critical commercial assays		
SPRIselect Reagent Kit	Beckman Coulter	Cat# B23318
Qubit dsDNA HS Assay Kit	Thermo Fisher Scientific	Cat# Q32854
High Sensitivity DNA Kit	Agilent	Cat# 5067-4626
KAPA Library Quantification Kit for Illumina Platforms	KAPA Biosystems	Cat# KK4824
EasySep™ RBC Depletion Reagent	STEMCELL Technologies	Cat# 18170
EasySep™ Human CD45 Depletion Kit II	STEMCELL Technologies	Cat# 17898
EasySep™ Human CD34 Positive Selection Kit II	STEMCELL Technologies	Cat# 17856
Chromium Next GEM Single Cell 3' GEM, Library & Gel Bead Kit v3.1, 16 rxns	10x Genomics	Cat# PN-1000121
Chromium Next GEM Chip G Single Cell Kit, 48 rxns	10x Genomics	Cat# PN-1000120
Dual Index Kit TT Set A, 96 rxns	10x Genomics	Cat# PN-1000215
NovaSeq 6000 S1 Reagent Kit v1.5 (100 cycles)	Illumina	Cat# 20012850
NovaSeq 6000 S2 Reagent Kit v1.5 (100 cycles)	Illumina	Cat# 20028316
Amicon® Ultra Centrifugal Filter, 50 kDa MWCO	Millipore	Cat# UFC505096
Oil Red O Staining Kit	LifeLine Cell Technology	Cat# LL-0052
High-Capacity cDNA Reverse Transcription Kit	Applied Biosystems	Cat# 4368814
Power SYBR Green PCR Master Mix	Applied Biosystems	Cat# 367659
Sample Kit for PhenoCycler-Fusion	Akoya Biosciences	Cat# 7000017
Antibody Conjugation Kit	Akoya Biosciences	Cat# 7000009
Deposited data		
Single-cell RNA sequencing raw and process data	This paper	NCBI GEO: GSE253355

REAGENT or RESOURCE	SOURCE	IDENTIFIER
CODEX spatial proteomics raw and processed data	This paper	https://doi.org/10.25452/figshare.plus.c.7174914
Vitessce interactive data explorer	This paper	https://cscb.research.chop.edu/index.php/bm-data
Experimental models: Cell lines		
Experimental models: Organisms/strains		
Oligonucleotides		
CODEX oligonucleotides, see Supplemental Table S5	Akoya Biosciences	various
qPCR oligonucleotides, see Supplemental Table S9	IDT	N/A
Recombinant DNA		
Software and algorithms		
Seurat v4 (Main Dataset) Seurat v5 (L/R Analysis)	https://satijalab.org/seurat/	Hao and Hao et al., <i>Cell</i> , 2021 Hao et al., <i>Nature Biotechnology</i> , 2023
Cellrangerv6.1.2	https://www.10xgenomics.com/support/software/cell-ranger/latest/analysis/inputs/cr-inputs-overview	N/A
PhenoCycler Experimental Designer v1.0.8	https://www.akoyabio.com/phenocycler	N/A
Fusion v1.0.6	https://www.akoyabio.com/phenocycler	N/A
Nolan Lab Neighborhood Analysis	https://github.com/nolanlab/NeighborhoodCoordination	Schurch et al., <i>Cell</i> , 2020
AUCell	https://github.com/aertslab/AUCell	Aibar et al., <i>Nature Methods</i> , 2017
CellChat v1	https://github.com/sqjin/CellChat	Suoqin Jin et al., <i>Nature Communications</i> 2021
CytoTRACE v0.3.3	https://cytotrace.stanford.edu/	Gulati et al., <i>Science</i> 2020
Spatstat	https://spatstat.org/	N/A
Wsireg	https://github.com/NHPatterson/wsireg	N/A
scRNA-Seq/CODEX Analysis Pipeline and Code	https://github.com/tanlabcode/spatial-bonemarrow-atlas	This paper
GraphPad Prism 10	https://www.graphpad.com/	N/A
Napari v0.4.14	https://napari.org/stable/	N/A
QuPath v0.4.3	https://qupath.readthedocs.io/en/0.5/	Bankhead, P. et al., <i>Scientific Reports</i> , 2017
Metascape v3.5	https://metascape.org/gp/index.html#/main/step1	Zhou et al., <i>Nature Communications</i> , 2019
FlowJo 10.8.1	https://www.flowjo.com/solutions/flowjo	N/A

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Mesmer 0.12.1	https://github.com/vanvalenlab/deepcell-tf/blob/master/notebooks/applications/Mesmer-Application.ipynb	Greenwald et al., Nature Biotechnology, 2021
µCT Evaluation Program v6.6	Scanco Medical, AG	N/A
Other		
Chromium Controller	10x Genomics	Cat# PN 120270
Phenocycler-FUSION	Akoya Biosciences	Cat# AB-000011
Eppendorf realplex ² Mastercycler	Eppendorf	Cat# 5345 017187
CFX Connect Real-Time PCR Detection System	BIO-RAD	Cat# 1855201
C1000 Touch™ Thermal Cycler	BIO-RAD	Cat# 1851197
NovaSeq 6000 System	Illumina	Cat# 20012850
2100 Bioanalyzer	Agilent	Cat# G2939A
10x Magnetic Separator	10x Genomics	Cat# 230003
EasySep™ Magnet	STEMCELL Technologies	Cat# 18000